

Immunobiochemical Reconstruction of Influenza Lung Infection - Melanoma Skin Cancer Interactions

Evgeni V. Nikolaev^{1,2}, Andrew Zloza^{3,4} and Eduardo D. Sontag^{5,6,*}

¹Center for Quantitative Biology, Rutgers University, Piscataway, NJ 08854, USA

²Clinical Investigations and Precision Therapeutics Program, Rutgers Cancer Institute of New Jersey, New Brunswick, NJ 08903, USA

³Section of Surgical Oncology Research, Division of Surgical Oncology, Rutgers Cancer Institute of New Jersey, New Brunswick, NJ 08903, USA

⁴Department of Surgery, Rutgers Robert Wood Johnson Medical School, New Brunswick, NJ 08903, USA

⁵Department of Electrical and Computer Engineering, and Department of Bioengineering, Northeastern University, Boston, MA 02115, USA

⁶Laboratory for Systems Pharmacology, Harvard Medical School, Boston, MA 02115, USA

Correspondence*:

Dr. Eduardo D. Sontag

e.sontag@northeastern.edu

2 ABSTRACT

3 It was recently reported that acute influenza infection of the lung promoted distal melanoma
4 growth in the dermis of mice. Melanoma-specific CD8+ T cells were shunted to the lung in
5 the presence of the infection, where they expressed high levels of inflammation-induced cell-
6 activation blocker PD-1, and became incapable of migrating back to the tumor site. At the same
7 time, co-infection virus-specific CD8+ T cells remained functional while the infection was cleared.
8 It was also unexpectedly found that PD-1 blockade immunotherapy reversed this effect. Here, we
9 proceed to ground the experimental observations in a mechanistic immunobiochemical model
10 that incorporates the T cell pathways that control PD-1 expression. A core component of our
11 model is a kinetic motif, which we call a PD-1 Double Incoherent Feed-Forward Loop (DIFFL),
12 and which reflects known interactions between IRF4, Blimp-1, and Bcl-6. The different activity
13 levels of the PD-1 DIFFL components, as a function of the cognate antigen levels and the given
14 inflammation context, manifest themselves in phenotypically distinct outcomes. Collectively, the
15 model allowed us to put forward a few working hypotheses as follows: (i) the melanoma-specific
16 CD8+ T cells re-circulating with the blood flow enter the lung where they express high levels of
17 inflammation-induced cell-activation blocker PD-1 in the presence of infection; (ii) when PD-1
18 receptors interact with abundant PD-L1, constitutively expressed in the lung, T cells render motility
19 loss. (iii) At the same time, virus-specific cells adapt to strong stimulation by their cognate antigen
20 by lowering the transiently-elevated expression of PD-1, remaining functional and mobile in the
21 inflamed lung, while the infection is cleared. The role that T cell receptor (TCR) activation and its
22 feedback play in the underlying processes is also highlighted and discussed. We hope that the
23 results reported in our study could potentially contribute to the advancement of immunological

approaches to cancer treatment and, as well, to a better understanding of a broader complexity of fundamental interactions between pathogens and tumors.

Keywords: influenza, melanoma, PD-1/PD-L1, incoherent feedforward loop, mathematical modeling

1 INTRODUCTION

It was recently reported that acute influenza A infection (A/H1N1/PR8) of the lung promoted distal B16-F10 skin melanoma growth in the dermis (Kohlhapp et al., 2016). It was also observed that melanoma-specific CD8+ T cells were shunted to the lung in the presence of the infection, where they expressed high levels of inflammation-induced cell-activation blocker PD-1, and became incapable of migrating back to the tumor site. At the same time, co-infection virus-specific CD8+ T cells remained functional while the infection was cleared. Finally, it was also unexpectedly found that blockade of PD-1 resulted in reversal of infection-mediated anti-tumor response disruption.

In this respect, it is very important to mention that the work by (Kohlhapp et al., 2016) was primarily motivated by two still unmet challenges: (i) emerging epidemiological studies reporting an increased cancer prevalence and cancer-specific deaths in patients with infections (Kohlhapp et al., 2016), and (ii) despite the fact that tremendous amount of work on immune response in the context of pathogenic co-infection has been done, findings in this field still remain discordant and a matter of debate, as also reviewed in (Kohlhapp et al., 2016; Zloza, 2018).

Motivated by the need to provide a more conceptual and quantitative biology insight into “the previously unrecognized acute non-oncogenic infection factor” accelerating tumor growth (Kohlhapp et al., 2016) and more broadly into the interactions between pathogens and cancer, and specifically, in order “to harness these interactions to improve microbial-based cancer therapy” (Zloza, 2018), we suggest a few immunobiochemical mechanisms and a simple mathematical model which may help to interpret the observed phenomena (Kohlhapp et al., 2016).

Our main results relate to two fundamental functional roles of immunity (Grossman and Paul, 1992, 2001, 2015): (i) adaptation of immune function, and (ii) competition between excitation and de-excitation (“push-pull”) factors possessing different response kinetics. In the context of this work, the loss of adaptation occurs in the expression of PD-1 receptors on anti-melanoma CD8+ T cells, a phenomenon that may constitute the essence of the previously unrecognized immunologic factor (Kohlhapp et al., 2016), while competing push-pull factors (Grossman and Paul, 1992) correspond to opposite outcomes of the corresponding kinetic motifs identified as incoherent feedforward loops (IFFLs) in the classification of (Alon, 2006). We briefly note that push-pull factors also play multiple fundamental roles in physiology (and biology) in general, *e.g.*, (Dampney et al., 2007).

Our working hypothesis is that the melanoma-specific T cells shunted to the lung in the presence of the infection express high levels of inflammation-induced cell-activation blocker PD-1, which upon interacting with PD-L1 constitutively expressed in the lung, render T cell motility paralysis (Zinselmeyer et al., 2013). At the same time, virus-specific cells adapt to strong stimulation by their cognate antigen by lowering the transiently-elevated expression of PD-1, remaining functional and mobile while the infection is cleared.

Although other important mechanisms may contribute to the previously unrecognized acute non-oncogenic infection factor (Kohlhapp et al., 2016), we focus our efforts on one concrete aspect of the problem, which is a gene regulatory network (GRN) that controls PD-1 expression. Indeed, the fact

that many other factors may contribute to the enormously complex molecular makeup of the acute non-oncogenic infection effect, such factors, obviously, do not exclude the interaction PD-1:PD-L1 playing a role as clearly seen from the data collected in (Kohlhapp et al., 2016). Thus, the importance of the PD-1:PD-L1 signal sent by the data cannot be disputed. Moreover, it is the PD-1:PD-L1 signal “detected” experimentally in (Kohlhapp et al., 2016) that defines the scope of our work aimed in uncovering relevant molecular detail in an unbiased way. We then develop and use a simple mathematical model in order to further illuminate the PD-1:PD-L1 role.

Specifically, a core component of our PD-1 gene-regulatory network (GRN) is a kinetic motif, which we call a Double Incoherent Feed-Forward Loop (PD-1 DIFFL), and which reflects known interactions between IRF4, Blimp-1, and Bcl-6 transcription factors (TFs). The different activity levels of the PD-1 DIFFL components, as a function of (a) the cognate antigen levels, (b) the T cell receptor (TCR) activity, and (c) the given inflammation context, manifest themselves in the T cell phenotypically distinct outcomes discussed in our work.

The rest of our work is organized as follows. In Sect. 2.1, the main results of (Kohlhapp et al., 2016) are briefly outlined. Alternative hypotheses potentially related to the unrecognized factor are discussed in Sect. 2.2. Here, the motivation for the development of the PD-1 DIFFL is also given. The PD-1 DIFFL is reconstructed in Sect. 2.3. We next attempt to falsify and validate the kinetic motif (PD-1 DIFFL) against key experiential observations in Sect. 2.4. The results of our mathematical modeling are described in Sect. 2.5. Finally, a literature review of the corresponding mechanistic detail, the model construction, and the model’s parameter justification can be found in Supplementary Information (SI).

2 RESULTS

We begin our analysis of the experimental data (Kohlhapp et al., 2016) by discussing a few alternative hypotheses, followed by the introduction of a number of mechanisms consistent with the discussed observations.

The selected mechanisms will then be formalized in terms of a relevant genetic molecular circuit (PD-1 DIFFL) that regulates PD-1 expression. Our proposed PD-1 DIFFL model is based upon molecular detail discovered previously, and is independent of the results obtained in (Kohlhapp et al., 2016).

We hope that the strong inference methodological approach (Platt, 1964) that guides our research will allow us to customize the PD-1 DIFFL to different inflammatory conditions (Kohlhapp et al., 2016) with the ultimate goal to capture both infection-tumor and infection-infection interactions at the mechanistic molecular level.

2.1 Linking observations with immunological mechanisms

A key challenge in the study of T cells within different dual immunological self (tumor) and non-self (infectious) contexts, is the organization of large amounts of relevant molecular and biochemical information (Sect. SI-1) compactly summarized in Table 1.

Table 1. A summary of the immunological reconstruction of infection-tumor interactions^a.

Observation	Description	Mechanism (hypothesis)
(O1) Anti-tumor CD8+ T cells are shunted to the infected site.	Tumor-specific CD8+ T cells of infected hosts were significantly reduced on day 6 in the TME compared to uninfected hosts and found at high levels at the site of infection but not observed in tissues unrelated to the tumor challenge or infection.	(O1-M1) Low-affinity immunological synapses formed between TCRs on anti-tumor CD8+ T cells and self-antigens on tumor cells lead to the lack of the Ag-induced arrest of the anti-tumor CD8+ T cells in the TME.
		(O1-M2) Infection-induced chemokines and cytokines amplify the tumor's ability to egress anti-tumor T _{EFF} from the TME.
		(O1-M3) Non-specific cardiovascular edema caused by infection-induced inflammation affects anti-tumor T _{EFF} trafficking.
		(O1-M4) Infection-induced chemokines chemoattract anti-tumor T _{EFF} to the infected lung.
		(O1-M5) Infection-induced IL-2 retains all types of T _{EFF} in the infected lung.
		(O1-M6) Infection-induced cytokines amplify expression of endothelial PD-L1, which in turn leads to paralysis of anti-tumor T _{EFF} in the inflamed lung due to high-affinity PD-1:PD-L1 bonds.
(O2) Cancer does not suppress the immune system anti-viral response which is capable at the same time of eradicating concomitant infections efficiently.	Cancer does not alter significantly the natural clearance of infection. Influenza infection also does not alter the natural clearance of VACV or the proportion of VACV-tetramer+ CD8+ T cells at the site of influenza infection.	(O2-M1) High-affinity immunological synapses formed between TCRs on anti-infection CD8+ T cells and nonself-antigens on infected cells lead to the Ag-induced arrest of the anti-infection CD8+ T cells inside the infected sites until a full clearance of the infection antigen.
(O3) Therapeutic PD-1 blockade reverses infection-mediated anti-tumor response disruption.	PD-1 blockade decelerates tumor growth in influenza-infected mice as well as rescues the percentage of anti-tumor CD8+ T cells within the TME.	(O3-M1) α PD-1 blockade shifts the dynamic equilibrium of the dynamically formed PD-1:PD-L1 bonds towards unbound forms of both PD-L1 and PD-1, allowing anti-tumor T _{EFF} to recover from immunologic paralysis and to gain motility.
		(O3-M2) Due to α PD-1 blockade, reactivated anti-tumor CD8+ T cells may recirculate back to the TME with constitutive lymph motion.
		(O3-M3) α PD-1 blockade reactivates PD-1-blocked signaling pathways leading to (i) improved killing capability, (ii) proliferation, (iii) suppression of PD-1 expression, (iv) protection against exhaustion, <i>etc.</i> in reactivated CD8+ T cells.

^{a)} Literature citations are directly inserted through the text (Sect. SI-1).

Specifically, Table 1 highlights the following key observations (O1) - (O5) made in (Kohlhapp et al., 2016):

- (O1) *Distant influenza-melanoma interaction*: Influenza-induced loss of anti-melanoma CD8+ T cells from the tumor micro-environment (TME) and their sequestration in the infected lung.
- (O2) *The host immune system's ability to respond to concomitant infection challenges*: influenza A virus (IAV) infection does not impede clearance of vaccinia virus (VACV) infection under the same conditions, nor tumor challenge changes the ability of the immune system to eliminate the infection.
- (O3) *Reactivation of exhausted (T_{EX}) anti-melanoma CD8+ T cells after anti-PD-1 (α PD-1) blockade*: (i) reactivated anti-melanoma CD8+ T cells which continue to reside in the TME regain their ability to contribute to the anti-tumor immune response and, additionally, (ii) reactivated anti-melanoma CD8+ T cells sequestered in the infected lung may regain their motility and return back to the TME, where they also aid in the anti-tumor response.
- (O4) *Reduced host survival*: Infection early in tumor formation reduces host survival by promoting tumor growth in the infected host.
- (O5) *Differential expression of PD-1 receptors by effector cells (T_{EFF}) in the infected lung*: Anti-melanoma CD8+ T cells express larger amounts of PD-1 receptors than anti-influenza CD8+ T cells under the same conditions in the infected lung.

2.2 From a physiologic systemic view of lymphocyte re-circulation to systems biology of PD1:PD-L1 interactions

It is known (Marchuk, 1997; Levin et al., 2016) that non-specific cardiovascular edema effects, caused by infection-induced inflammation in the infected site, redirect the blood-flow to the site of infection-induced inflammation. Therefore, it is highly appealing to explain the observed accumulation of anti-melanoma CD8+ T cells in the infected lungs, (O1), by non-specific inflammation effects only.

Note that the lymphocyte recirculation routes are phenotype-dependent and significantly differ for naïve/memory/effector subsets (Owen et al., 2013). We leave the corresponding details specific to the different subsets out of the discussion that follows. What is relevant to our work is that all newly activated cytotoxic T lymphocytes (CTLs) exit the corresponding lymph nodes into lymph via lymphatic ducts before they enter circulation via the great veins, and then flow through the pulmonary circulation (Fig. 1). Under resting non-inflamed conditions, re-circulation of lymphocytes between lungs and blood is very rapid, with the average residence time in the lungs less than *one minute* (Ganusov and Auerbach, 2014).

After leaving the heart and lungs, the traveling CTLs continue to flow into systemic circulation, followed by their ultimate but not instantaneous homing in the corresponding infectious or tumor sites. Indeed, lymphocytes on average must pass via vasculature of the lung or liver about 10 times or even more times (Poleszczuk et al., 2016) before they migrate to one of the secondary lymphoid tissues (Owen et al., 2013)[BOX 14.2]. For example, it was shown that if anti-tumor CTLs were activated in the breast, they would perform on average about eight circulatory transient cycles before extravasation into the tumor site (Poleszczuk et al., 2016).

Before reconciling the experimental observation (O1) with these studies, we have to briefly discuss a unique role that the lung plays in the physiology and immunology of trafficking lymphocytes under both resting and inflamed conditions.

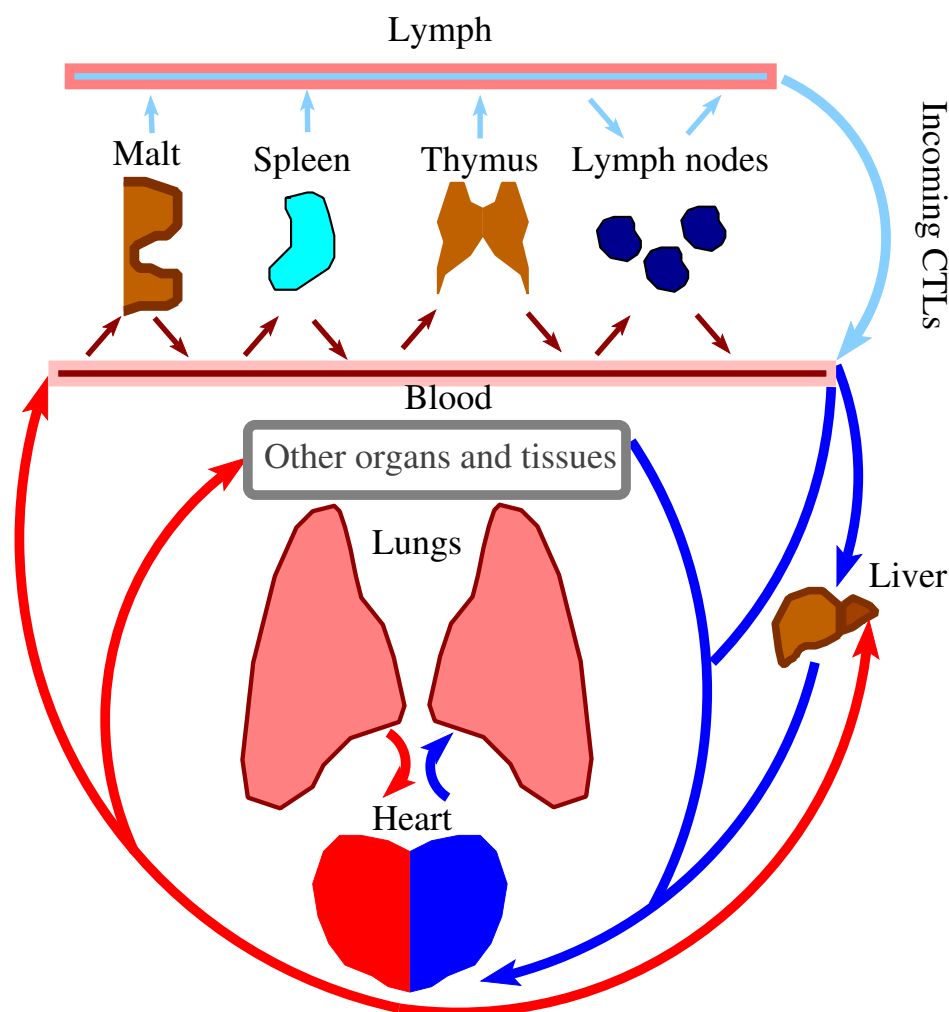


Figure 1. Schematic representation of lymphocyte re-circulation routes. There are two different routes for naïve and activated trafficking lymphocytes (Ford and Gowans, 1969; Owen et al., 2013). First, due to the data discussed in (Owen et al., 2013, Ch.14) and, independently estimated in (Van den Berg, 2011, p.23) after approximately 30 min. transit time in the blood, about 45% of all naïve lymphocytes (produced by the thymus and bone marrow) migrate to the spleen, where they reside for about 5 hrs. Another 45% of lymphocytes enter various peripheral nodes, where they remain for 12 to 18 hrs, scanning stromal cell surfaces. A smaller fraction of lymphocytes migrate to secondary lymphoid tissues (skin, gastrointestinal, etc.), to protect the organism against the external environment. Thus, about 5% of the lymphocytes are, at resting conditions, in the blood, and the majority resides in the lymph nodes. Second, as discussed in (Poleszczuk et al., 2016) activated CTLs enter the blood system via the great veins, flow through the pulmonary circulation, and, then, continue into systemic circulation. Venous blood from gastrointestinal tract and spleen goes to the liver through the hepatic portal vein. In all cases, lymphocytes migrate from the blood into lymph nodes through high-endothelial venues, specialized areas in postcapillary venues. *Remarks.* (a) MALT is Mucosa Associated Lymphoid Tissue. (b) Lymph nodes have both afferent and efferent lymphatic vessels, while MALT, Spleen, and Thymus have efferent lymphatic only (Owen et al., 2013).

137 Experimental studies have revealed that different subsets of lymphocytes, including naïve/memory/activated
 138 effector T cells, transiently accumulate in the lungs (Toapanta and Ross, 2009; Odoardi et al., 2012) both by
 139 means of and, what is also extremely important, without specific antigen-dependent recruitment of CTLs to
 140 the lung (Anderson et al., 2012). Anderson et al. (2012) further discuss “numerous observations indicating
 141 that T cell trafficking within the lung is starkly different from what is known about T cell trafficking in

most nonlymphoid tissues,” including the fact that lymphocyte extravasation into the lung is chemokine independent (Klonowski et al., 2004; Bromley et al., 2008). So, one must revisit the observation (O1) by taking into account the unique role that the lung may play in lymphocyte retention even in the absence of influenza A related antigen-induced chemokine gradients that would additionally force anti-melanoma CTLs to extravasate into the lung epithelium, should influenza A infection be present.

Unfortunately, the above results and the unique role of the lung to transiently retain lymphocytes still do not explain the difference in the observations (O1) and (O2), nor they explain the observation (O3), for the following reasons.

First, concerning the observations (O1) and (O2), both anti-melanoma and anti-infection CTLs should follow the same pattern of multiple vascular re-circulation cycles as discussed above (Fig. 1). However, under similar re-circulation patterns, the presence of IAV infection impedes tumor clearance, while, at the same time, both IAV and another concomitant infection (*e.g.*, VACV) are cleared efficiently as one infection would be cleared in the absence of another. Specifically, the question “Why are anti-melanoma CTLs retained in the infected lung, while anti-VACV infection CTLs are not?” remains unanswered.

Given the large literature body on the importance of PD-1 receptors in immune response and the observations (O3) and (O5), we decided to explore theoretically whether molecular signaling pathways initiated by PD-1 ligated with PD-L1 would provide at least one plausible mechanism to explain the results obtained in (Kohlhapp et al., 2016).

We have excluded PD-L2 from our model and only consider PD-L1 in the analysis that follows. Indeed, PD-L2 has restricted expression on macrophages, dendritic cells (DCs), and mast cells, while PD-L1 is expressed more broadly in order to mediate T cell tolerance in non-lymphoid tissues (Owen et al., 2013; Alsaab et al., 2017). Besides, mathematical simulations based on the biophysical and expression data have revealed an unexpectedly limited contribution of PD-L2 to PD-1 ligation during interactions of activated T-cells with APCs (Cheng et al., 2013).

To this end, the immune system has apparently evolved the inhibitory PD-1/PD-L1 pathway as a result of the need to control the degree of inflammation at locations expressing the antigen in order to secure normal tissue from damage and also to maintain peripheral tolerance (Grossman and Paul, 2001; Alsaab et al., 2017). This includes the constitutive expression of PD-L1 in large quantities in various tissues such as lungs, pancreatic islets, ovary, colon, *etc.* (Sakaguchi et al., 2008; Fife et al., 2009; Francisco et al., 2010; Galante et al., 2012; Schietinger and Greenberg, 2014; Bardhan et al., 2016) by which cross-reactive effectors that survive positive selection are also muted to maintain the peripheral tolerance (Zloza, 2018).

2.3 Incoherent feed-forward regulation of PD-1 expression

PD-1 expression on CD8+ T cells is known to be regulated at the level of transcription of its gene *pdccl1* (Lu et al., 2014). More precisely, two upstream conserved regulatory CR-B and CR-C regions (Lu et al., 2014) are key for PD-1 expression in response to CD8+ T cell activation. Specifically, TCR signaling induces PD-1 gene expression through the transcriptional activator, Nuclear Factor of Activated T cells, cytoplasmic 1 (NFATc1) (Fig. 2), which binds to CR-C after translocation to the nucleus (Collins and Henderson, 2014; Lu et al., 2014).

Next, the down-regulation of PD-1 during acute infection (Wherry et al., 2007) suggests that there exists a mechanism that directly represses its expression after initial activation events. Indeed, Blimp-1 (B Lymphocyte-Induced Maturation Protein 1) (Martins and Calame, 2008) has been found to be induced during the *later* stages of CD8+ T cell activation and was shown to be required for the efficient terminal

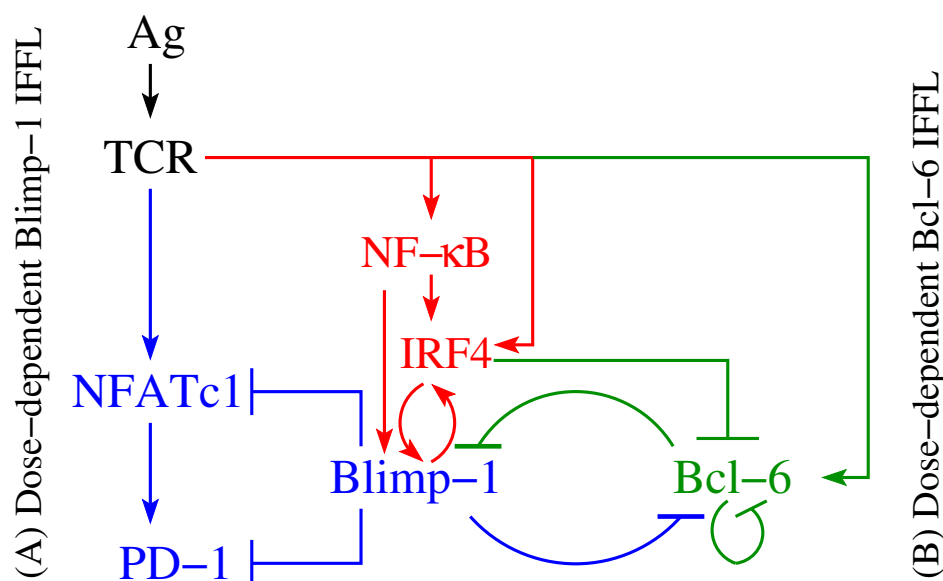


Figure 2. Regulation of PD-1 expression. Two different IFFLs, sharing a common set of species and regulatory activities highlighted in red, are presented. Both IFFLs are activated by the same input (Ag). The left hand side panel (A) depicts a dose-dependent biphasic activation of PD-1. The elements of the corresponding IFFL are highlighted in blue and red colors. When the input, the Ag dose, increases, the output, the PD-1 level, first also increases but then subsequently decreases. The right hand side panel (B) corresponds to a dose-dependent activation of Bcl-6. The elements of the corresponding IFFL are highlighted in green and red colors. Over a certain range of input dose, the Ag level, the output, in this case Bcl-6 level, increases but with a subsequent increase in the Ag dose, the Bcl-6 level then decreases.

184 differentiation of effector CD8+ T cells (Lu et al., 2014). When Blimp-1 is suppressed, the same data
 185 suggest that in the absence of Blimp-1, PD-1 expression is maintained by NFATc1 (Fig. 2).

186 For the sake of completeness, we recall that the existing data also suggest that Blimp-1 represses PD-1
 187 gene expression in CD8+ T cells using three distinct molecular mechanisms (Lu et al., 2014):

- 188 (1) regulation of the expression of PD-1's activator NFATc1;
- 189 (2) alteration of the local chromatin structure; and
- 190 (3) eviction of the activator NFATc1 from its site that controls PD-1 expression.

191 In addition, Blimp-1 has been found to be a transcriptional antagonist of proto-oncogene Bcl-6 (B cell
 192 lymphoma 6 transcription factor), and vice versa (Fig. 2) (i.e., Blimp-1 and Bcl-6 are known to mutually
 193 repress one another) (Calame, 2008; Sciammas et al., 2011; Singh et al., 2014; Daniels and Teixeira, 2015;
 194 Speiser et al., 2016). Specifically, Blimp-1 can bind to the *bcl6* promoter (Shaffer et al., 2002).

195 Although it is not known exactly how Bcl-6 inhibits Blimp-1 in T cells, it is well known that in B cells
 196 Bcl-6, in association with a corepressor MTA3, represses *prdm1* by binding to sites in *prdm1* intron 5 and
 197 intron 3 (Shaffer et al., 2000; Calame, 2008; Pasqualucci et al., 2011). We take this fact into consideration
 198 because signaling pathways and their activation are similar in both B and T cells (Owen et al., 2013).
 199 Additionally, Bcl-6 binds its own promoter and inhibits its own transcription (Fig. 2), thus implementing
 200 an autoregulatory loop (Pasqualucci et al., 2003; Martinez et al., 2012) (Fig. 2).

201 Competing with Bcl-6 for intron 5 in *prdm1*, IRF4 (Interferon Regulatory Factor) (Sciammas et al., 2006;
 202 Shaffer et al., 2009; Man et al., 2013; Iwata et al., 2017) is shown to be a direct activator of *prdm1* (Fig. 2)

by binding to a site in intron 5 (Calame, 2008). At the same time, IRF4 directly represses gene *bcl-6* by binding to a site within its promoter (Calame, 2008; Shaffer et al., 2009), which is rich in IRF4-binding sites (Martinez et al., 2012).

Because IRF4 is known to be activated both directly via TCR and by NF- κ B (Boddicker et al., 2015; Vasanthakumar et al., 2017), we have then sought to determine who activates NF- κ B in this context and found that NF- κ B is activated by TCR signaling (Calame, 2008; Ahmed and Nandi, 2011; Paul and Schaefer, 2013; Daniels and Teixeira, 2015). Several potential NF- κ B binding sites in the *prdm1* promoter have been suggested. It is also known that IRF4 can bind to its own promoter, supporting a positive feedback mechanism by which high IRF4 expression can be maintained (Shaffer et al., 2008; Martinez et al., 2012).

There are additional signaling routes leading to the activation of IRF4 (*e.g.*, via Akt-mediated pathways) which are not discussed here (Calame, 2008).

After a careful analysis of the reconstructed molecular interactions, we have come to the conclusion that this intricate reaction network consists of two subnetworks (Fig. 2). Both subnetworks have the same input from the activated TCR, while the outputs of the subnetworks are different. Namely, PD-1 is the output of the subnetwork color-coded in blue and red, while Bcl-6 is the output of the subnetwork color-coded in green and red. The two subnetworks share a number of common species and interact with one another via repressive interactions mediated by the three key species color-coded in red, (*i*) IRF4, (*ii*) Blimp-1, and (*iii*) Bcl-6.

Each of the two subnetworks corresponds to a gene-regulatory network (GRN) motif known as an incoherent feed-forward loop (IFFL) (Alon, 2006). Because the PD-1 circuit is formed of two such IFFLs, we call it a Double Incoherent Feed-Forward Loop (DIFFL).

Our IFFL network may be viewed as a mechanistic instantiation of a conceptual signal discrimination model based on a competition between “excitation” and “de-excitation” factors possessing different response kinetics, as initially introduced by Grossman and Paul (1992). The latter concept has been gradually applied successfully in multiple studies since 1992 as reviewed in (Grossman and Paul, 2015). In that sense, we address with our model the following goal formulated in (Grossman and Paul, 1992): “More explicit rules of organization, or models, need to be explored. Such rules should suggest, in particular, how the functional segregation of immunological responses may reasonably come about.”

2.4 PD-1 expression within different inflammatory contexts

We next attempt to validate the PD-1 DIFFL motif (Fig. 2) against all observations reported in (Kohlhapp et al., 2016) by following the falsification and validation methodology (Oreskes et al., 1994), which is also fundamental to any modeling study. Figure 3 will be instrumental in our analysis that follows.

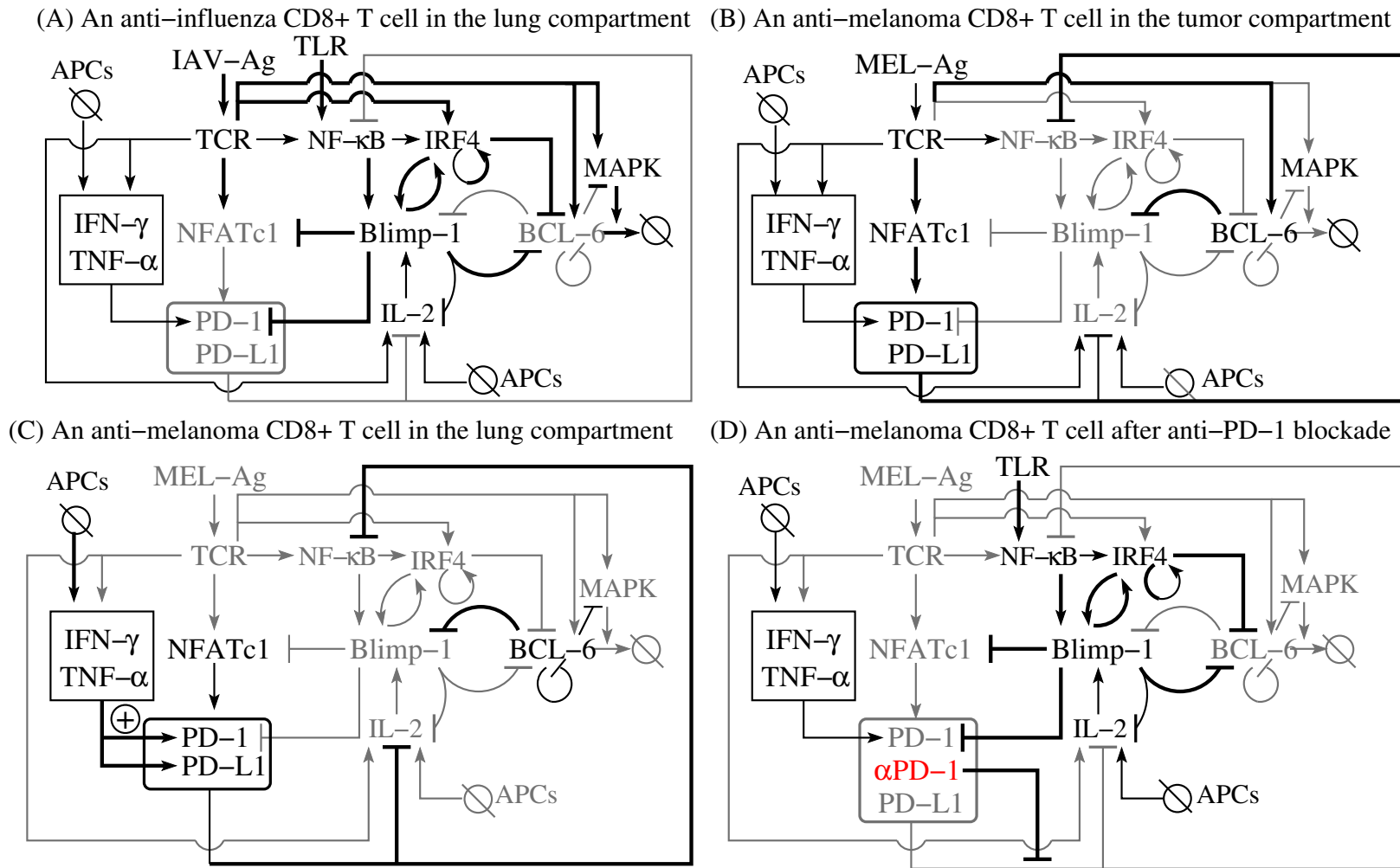


Figure 3. The PD-1 DIFFL motif in the context of complex influenza-tumor interactions. Panel (A) shows the PD-1 DIFFL response in an anti-influenza CD8+ T cell in the infected lung. Panel (B) shows the response of the PD-1 DIFFL circuit in an anti-tumor CD8+ T cell in the TME. Panel (C) shows the PD-1 DIFFL response in an anti-tumor CD8+ T cell in the influenza-infected lung. Panel (D) shows the PD-1 DIFFL response in an anti-tumor CD8+ T cell in the influenza-infected lung after PD-1 blockade. Gray color corresponds to weak or disabled reactions shaped by the given inflammation context. Symbol + inside a circle in panel (C) shows the additional PD-1 activation route initiated by external cytokines in the case when the Blimp-1 mediated repression of PD-1 expression is absent. This route does not play any significant role in the case when the expression of PD-1 is suppressed by active Blimp-1 as in panel (A). Arrows denote activation, and barred lines denote repression. The abbreviation APCs stands for (influenza) Antigen Presenting Cells.

Panel (A) of Fig. 3 shows a biochemical reaction network reconstruction customized for the case of an anti-influenza cytotoxic effector T cell, T_{EFF} , in the presence of large amounts of cognate Ag in the infected lung. In this case, the immunological complexity of interactions involving cytokines is already overwhelming (Roshani et al., 2014; Dittrich et al., 2015; Grossman and Paul, 2015; Duvigneau et al., 2016; Minn and Wherry, 2016; Odenthal et al., 2016). For example, IL-2 activates and is simultaneously repressed by active Blimp-1 both directly and indirectly (Martins et al., 2008; Collins and Henderson, 2014).

The abundance of the cognate viral Ag in the infected lung leads to a strong TCR activation which, in turn, results in the simultaneous activation of Blimp-1 and degradation of Bcl-6 (Sect. 2.3) followed by suppression of PD-1 transcription with its subsequent degradation. The biochemical detail can explain transient and rapid PD-1 expression followed by downregulation of PD-1 expression in the presence of acute infection (Wherry et al., 2007), see also Sect. SI-1.2.

All this may also explain why anti-infection CD8+ T cells are not exhausted during the first phase of the biphasic response of the PD-1 DIFFL-circuit (Sect. 2.3) despite the fact that bystander and tissue cells express large amounts of PD-L1 caused by large concentrations of pro-inflammatory cytokines such as $\text{INF}\gamma$ (SI-1.1). Recall that large amounts of PD-L1 are already constitutively expressed in the lung under resting condition in the absence of any infection (Sect. 2.3).

Panel (B) of Fig. 3 shows the response of the reconstructed circuit in the tumor microenvironment (TME). Specifically, anti-melanoma CD8+ T cells overexpress PD-1 in the presence of large amounts of tumor-specific cytokines such as IL-6, a well-described regulator of Bcl-6 expression (Speiser et al., 2016). Due to relatively low levels of tumor Ag and a weak self-Ag TCR signal (Richards et al., 2016) of anti-tumor CD8+ T cells, the TCR is not activated strongly enough to activate Blimp-1 and, at the same time, the weak activation of the TCR sets the first phase of the biphasic response of the dose-dependent PD-1 DIFFL motif (Fig. 2).

Indeed, the PD-1 DIFFL strongly activates Bcl-6 for small and medium TCR strengths, and weakly activates Bcl-6 for high activity levels of TCR. As a result, Bcl-6 is overexpressed, while Blimp-1 is not expressed in the melanoma TME (Speiser et al., 2016), which leads to the overexpression of PD-1 on the surface of anti-tumor CD8+ T cells.

Panel (C) of Fig. 3 shows the PD-1 DIFFL in an anti-melanoma T_{EFF} relocated into the infected lung. In this case, the conditions discussed just above to introduce panel (B) play the role of a spark plug that activates the transcription of Bcl-6, which represses *prdm1* even after the relocation of the anti-tumor T_{EFF} into the lung.

These relocated T_{EFF} can now sense the elevated levels of $\text{INF}\gamma$ and $\text{TNF}\alpha$, which are abundant in the infected site, and which are produced by professional antigen presenting cells (APCs) (Sect. 2.2).

The cytokines strongly stimulate the expression of both PD-1 and PD-L1 (Liu et al., 2017b), as well as maintain the expression of PD-1 on the surface of the anti-melanoma T_{EFF} , initially sparked by the ligation of TCRs with cognate tumor Ags during the time when the T_{EFF} cells were present in the TME before their relocation to the lung.

Because the tumor Ag is absent from the infected lung, the TCR is not ligated, and, hence, all routes leading to the activation of both Blimp-1 are disabled. We can thus propose that the major route contributing to PD-1 overexpression here is mediated by $\text{INF}\gamma$ and $\text{TNF}\alpha$. The corresponding PD-1 expression activation route is marked by sign + inside a circle in panel (C) of Fig. 3.

Recall that large quantities of PD-L1 are constitutively expressed in the lung already under resting conditions (Sect. 2.2). PD-1 mediated control of immune responses depends on interactions between PD-1 on CD8⁺ T cells and PD-L1 in tissues (Nirschl and Drake, 2013). Importantly, such PD-1:PD-L1 interactions can result in CD8⁺ T cell motility paralysis (Zinselmeyer et al., 2013; Schietinger and Greenberg, 2014; Stein et al., 2016).

We introduce the paralysis mechanism (Fig. 4) in detail in (O1-M6) (Sect. SI-1.3) and believe that this mechanism can provide a valuable insight into the previously unrecognized factor contributing to the retention of anti-melanoma CD8⁺ T cells shunted to the influenza A infected lung (Kohlhapp et al., 2016). Of course, other yet unknown mechanisms may exist and need to be elucidated in order to provide a more complete explanation of the retention effect (Kohlhapp et al., 2016). Therefore, additional experimental observations should be obtained.

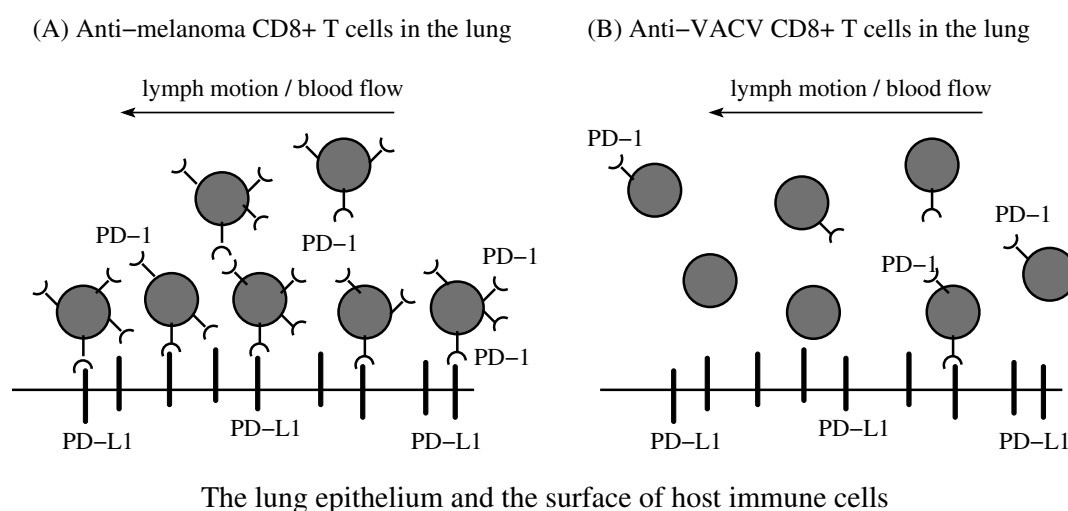


Figure 4. PD-1:PD-L1 induced paralysis of the anti-tumor exhausted CD8⁺ T cells in the infected site. Panel (A) suggests that anti-melanoma T_{EFF} cells become paralyzed in the infected lung. In contrast, panel (B) suggests that anti-VACV T_{EFF} studied in (Kohlhapp et al., 2016) can freely enter and leave the infected lung with the lymph motion and blood flow due to the lack of large amounts of PD-1 receptors on their surface. The immune suppressive environment (Grossman and Paul, 2001) induced by inflammation in the infected lung is caused by multiple interactions between PD-1 receptors, expressed in large quantities on the surface of the anti-melanoma T_{EFF} , and the PD-L1 ligands expressed in large quantities on the surface of various host immune cells (macrophages, DCs, and MDSCs) and the epithelium (Bardhan et al., 2016).

288

The study conducted by Cheng et al. (2013) reports that “it now seems that very stable complexes are not prerequisite for potent inhibitory PD-1:PD-L1 signaling” because measurements of the human and mouse PD-1 binding to PD-L1 affinities suggest that potent inhibitory signaling can be mediated by weak interactions.

Zinselmeyer et al. (2013) further stress: “Prolonged motility arrest is an excellent host strategy to decrease T cell efficiency and likely facilitates exposure to multiple regulatory pathways. PD-1:PD-L1 blockade is known to restore function to virus-specific and tumor-specific T cells, and has shown some promise in recent clinical trials.”

Although dissociation and association of the complex PD-1:PD-L1 are assumed to be fast (Lai and Friedman, 2017; Nikolopoulou et al., 2018), this however does not preclude the long-known loss of T cell mobility due to multiple PD-1:PD-L1 interactions (Pauken et al., 2016; Wang et al., 2017).

Panel (D) of Fig. 3 shows the PD-1 DIFFL in an anti-melanoma T_{EFF} cell in the infected lung after administration of PD-1 (α PD-1) blockade. Recall that the NF- κ B pathway is downregulated in exhausted CD8+ T cells (Speiser et al., 2016). To this end, the PD-1 blockade (marked by symbol α PD-1 color-coded in red) in Fig. 3(D), removes the brake (Liechtenstein et al., 2012) from the corresponding T cell signaling pathways (see Sect. 2.1, observation O(3), and Table SI-I.1) leading to overexpression of NF- κ B (Pauken et al., 2016; Siefker-Radtke and Curti, 2018). Additionally, NF- κ B activation is positively regulated through TNFR (TNF Receptor) and TLR (Toll-like Receptor) sensing TNF α and viral materials in the infected lung, respectively (Kumar et al., 2004; Liu et al., 2017a; Sun, 2017).

As discussed earlier, NF- κ B activates IRF4 (Calame, 2008), and the latter directly represses Bcl-6 (Calame, 2008). In turn, the repression of Bcl-6 removes the brake from the overexpression of Blimp-1, which then leads to reduced numbers of PD-1 receptors on the surface of reactivated anti-melanoma effector cells. This may allow the reactivated T_{EFF} to become mobile (Table SI-I.1) with a potential to relocate back to the melanoma TME with the lymph flow and blood circulation as discussed in the mechanism (O1-M6). Indeed, it is well known that after the T_{EFF} re-circulation in the blood (Poleszczuk et al., 2016), effector T cells are preferentially found in the lymph nodes in which their activation occurred, and in the area drained by those lymph nodes (Westermann et al., 2001).

The above conclusions are also based on the experimental evidence that PD-1:PD-L1 interactions contribute to reduced T cell motility on day 7, and therapeutic blockade of PD-1:PD-L1 restore CD8+ T cell motility within 30 min (Zinselmeyer et al., 2013). Although we use the references (Zinselmeyer et al., 2013; Stein et al., 2016) in order to support our hypotheses, additional experimental research is needed to understand deeper the paralysis phenomenon (Schietinger and Greenberg, 2014; Stein et al., 2016).

We conclude our discussion of the PD1 DFFIL motif by noting that the core of the reconstruction (Fig. 2) fits well to all discussed inflammatory contexts (Fig. 3).

2.5 Probing immunobiochemical reconstruction modeling

Our modeling goal here is quite simple. Given the discussed specificity of PD-1 expression (Sect. 2.4) with respect to different amounts of antigen available in the medium and different values of TCR affinity in terms of the values of the off-rate constant k_{off} for the Ag:TCR bond (Lever et al., 2016; Rendall and Sontag, 2017), we focus on the analysis of the dependence of the levels of key species, Bcl-6, IRF4, Blimp-1, and PD-1, on the two parameters, (i) the antigen concentration, Ag, and (ii) the values of k_{off} defined in Sects.-SI-2 and SI-3.

2.5.1 Modeling PD-1 expression in the absence of PD-L1

We first consider the case when the PD-1:PD-L1 interaction is absent from the model by setting $\phi_L(P) \equiv \Phi(P) \equiv 1$ corresponding to the condition $L = [PD-1:PD-L1] = 0$ in both equations (SI-2.1c) and (SI-2.2a).

Typical plots for the (non-dimensional) steady-state (Khibnik et al., 1993) concentration levels of PD-1, Bcl-6, Blimp-1, and IRF4 in the absence of the PD-1:PD-L1 interaction and at the different values of k_{off} are shown in Fig. 5. The model's nondimensionalization is done in Sect. SI-2 and SI-3.

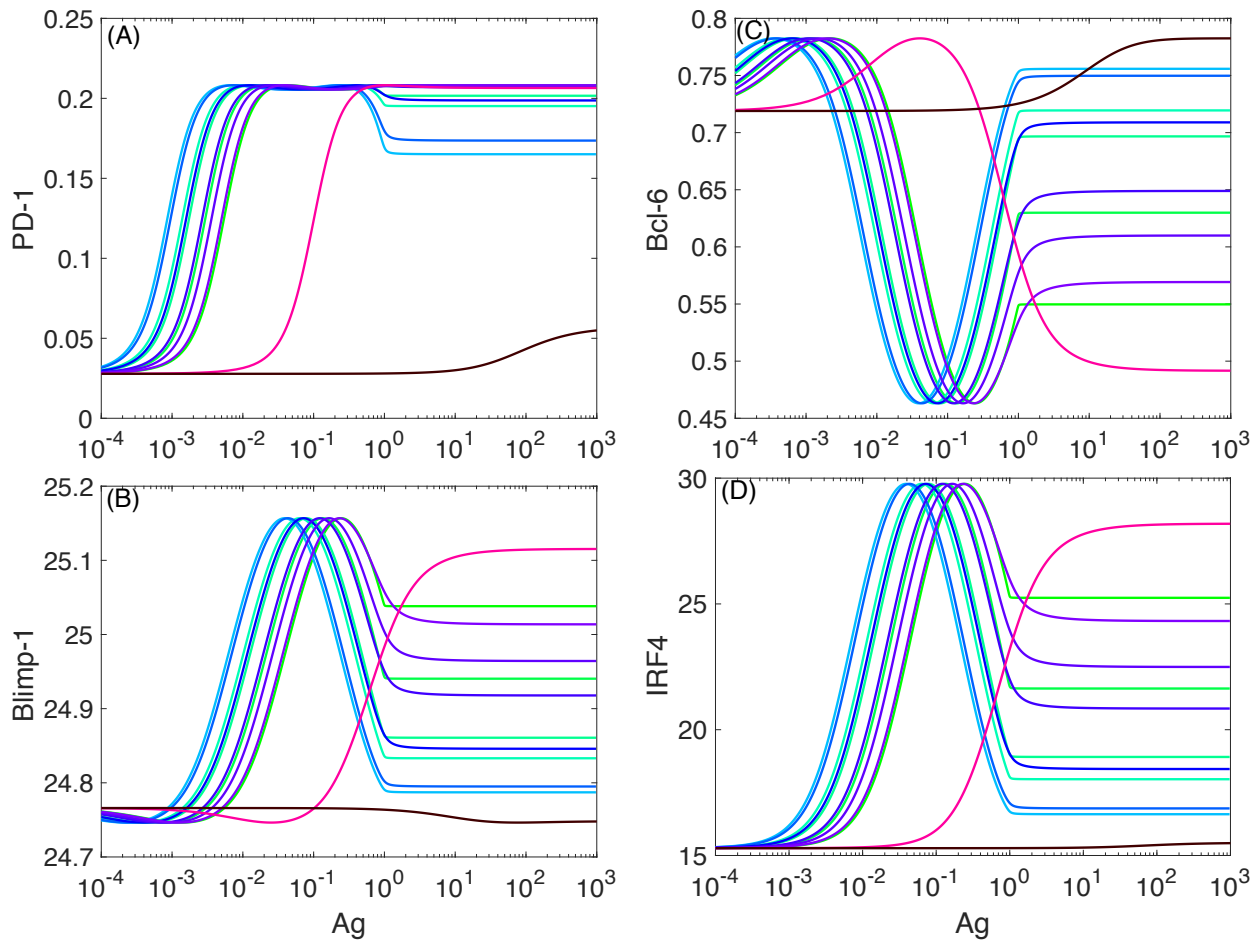


Figure 5. PD-1 DIFFL responses in the absence of PD-1:PD-L1 interaction. The color-coded plots corresponds to the PD-1 DIFFL-induced adaptation with respect to increasing Ag-levels. To obtain a full adaptation, approximately a 10^3 -fold increase in the Ag-level is required. Four different (bottom-up) shades of green color correspond to $k_{\text{off}} = 10^{-4}$, 2.03×10^{-4} , 4.13×10^{-4} , and 5.88×10^{-4} , respectively. Two shades of blue color correspond to $k_{\text{off}} = 2.43 \times 10^{-3}$ and 7.01×10^{-3} , respectively. Four (top-down) shades of purple color correspond to $k_{\text{off}} = 2.03 \times 10^{-2}$, 4.13×10^{-2} , 5.88×10^{-2} , and 8.38×10^{-2} . Magenta color corresponds to $k_{\text{off}} = 1.0$. Black color corresponds to $k_{\text{off}} = 49.24$.

337 We next discuss the case of small values of k_{off} from the set of the values given in the legend of Fig. 5.
 338 We observe from Fig. 5 that the level of PD-1 (panel (A)) becomes rapidly elevated already at very small
 339 values of the scaled Ag-concentration (Sect. SI-1). A further increase in the scaled Ag-concentration results
 340 in the formation of the PD-1 level plateau, followed by a drop in PD-1 levels.

341 The increase in the level of PD-1 (Fig. 5(A)) is fully aborted when the level of Blimp-1 (panel (C))
 342 reaches the threshold sufficient to suppress PD-1 expression initiated by TCR activation. We interpret the
 343 top (left) plateau in the level of PD-1 (Fig. 5(A)) as corresponding to the homeostasis maintained by both
 344 the PD-1 DIFFL and the negative feedback activation of TCR which we discuss shortly below. At the same
 345 time the bottom (right) plateau in the level of PD-1 (Fig. 5(A)) can be interpreted as an adaptation to high
 346 levels of Ag (Grossman and Paul, 1992), a direct consequence of adaptive properties of IFFLs (Alon, 2006;
 347 Kim et al., 2008; Shoval et al., 2011; Chiang et al., 2014; Skataric et al., 2015; Sontag, 2017; Rahman et al.,
 348 2018).

We further observe that in complete agreement with the theory of IFFLs demonstrating biphasic steady-state behavior (Alon, 2006; Kim et al., 2008; Shoval et al., 2011), the levels of Blimp-1 and IRF4 first increase and then decrease, and, at the same time, the level of Bcl-6 first decreases and then increases, while the level of Ag is constantly increased. Remarkably, the levels of all the three species almost perfectly adapt to their respective original states formed initially at very low levels of Ag, when the level of Ag becomes high enough to establish adaptation. A similar adaptive phenotype is discussed using an example of a generalized enzyme network in (Chiang et al., 2014).

Consider now the case of large values of k_{off} from the set of the values given in the legend of Fig. 5. In this case, the response of the PD-1 DIFFL becomes abnormal, when all remarkable adaptive properties are completely lost. Even in the case of a very large value of k_{off} , the model predicts a tonic expression of PD-1 corresponding to very small nonzero values coded in black color in (Fig. 5(A)). We believe that this tonic expression of small PD-1 levels can be attributed to the immune tolerance discussed in Sect. SI-1.

To better see the role of IFR-4 and its impact on the level of PD-1, we then completely disabled IRF4 by setting the value of the parameter k_b to zero, $k_b = 0$ in the equation (SI-3.1d). This computational experiment can be thought of as an “*in silico* IRF4-knockout.” The corresponding plot of PD-1 levels against the Ag-concentration is shown in Fig. 6(A).

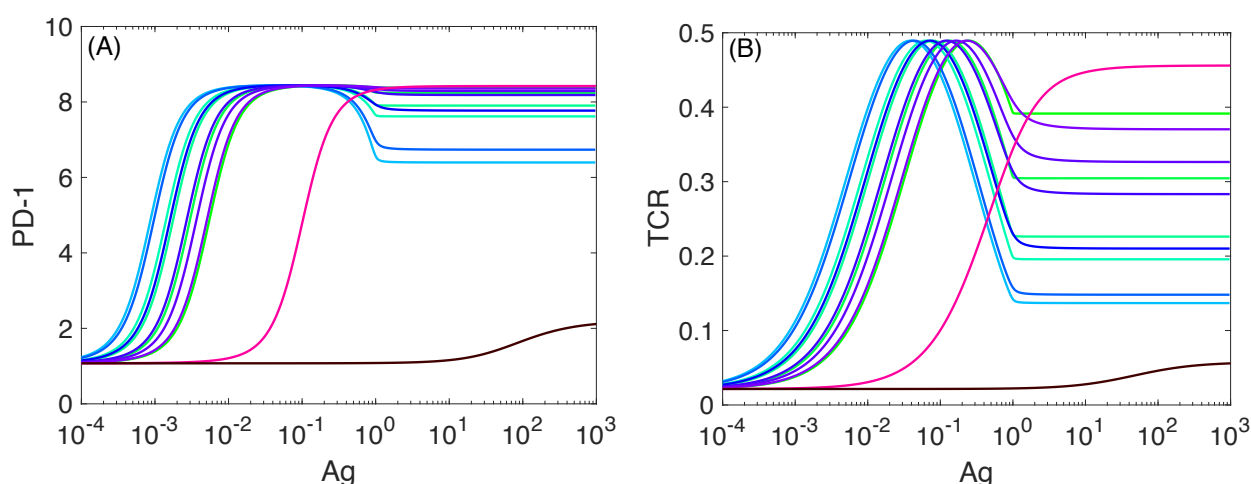


Figure 6. Expression of PD-1 in the case when the expression of IRF4 is disabled. The level of TCR activity in panel (B) is computed from the model developed in (Lever et al., 2016) as explained in SI-2. All other explanations and parameter values are as in Fig. 5.

364

Surprisingly, the shapes of all PD-1 level plots obtained for the same set of k_{off} values as in Fig. 5 are preserved, and only the magnitudes of the corresponding levels are changed by a factor of 40 or more.

Motivated by these computational predictions, we checked if IRF4 knockout results were previously reported in the literature and found that *irf4*-deficient CD4+ T cells display increased expression of PD-1 associated with T cell dysfunction (Wu et al., 2017a,b). However, the role of IRF4 is still poorly understood as it can be completely opposite in the cases of acute and chronic infections (Chennupati and Held, 2017; Wu et al., 2017a).

The second interesting observation (Fig. 6(B)) is that while the PD-1 DIFFL regulatory function is lost due to *in silico* knockout of IRF4, the adaptation of PD-1 expression with respect to Ag levels (Fig. 6(A)) is still preserved by the negative feedback regulation of TCR activity (Fig. 6(B)) (Štefanová et al., 2003;

375 François et al., 2013; Grossman and Paul, 2015; Watson et al., 2016). Both the TCR activation and the
 376 negative feedback are interpreted as another IFFL in (Lever et al., 2016). Collectively, we can thus conclude
 377 that the PD-1 transcription and its adaptation to high levels of antigen is regulated by multiple feed-forward
 378 loops.

379 2.5.2 Modeling PD-1 expression in the presence of PD-L1

380 We observe that in the presence of PD-1:PD-L1 interactions, the maximum levels of PD-1 and Bcl-6
 381 increase (by a factor of 6.75 and 7.86 respectively, but, of course, these numbers are only meaningful in
 382 our model and with the parameters used, and they do not have biological significance). (Fig. 7). At the
 383 same time, the levels for Blimp-1 and IRF4 are negligibly small, which allows us to interpret that the
 384 transcription of these two species is almost fully suppressed (Fig. 7).

385 From our comparison of the PD-1 level plots in Fig. 6 and Fig. 7, we can conclude that the PD-1:PD-L1
 386 interaction plays the role of an amplifier of transient activation of PD-1 transcription, initiated by the
 387 ligation of TCR with Ag presented with an MHC (Sect. SI-1).

388 PD-1:PD-L1 interactions may terminate signal transduction pathways, including those pathways that lead
 389 to the activation of IRF4 and Blimp-1, by recruiting phosphatases (Liechtenstein et al., 2012; Ohaegbulam
 390 et al., 2015; Arasanz et al., 2017).

391 Our last computational experiment compares quantitatively the PD-1 level on the surface of an anti-
 392 melanoma CD8+ T cell shunted to the lung with the PD-1 level on the surface of an anti-influenza CD8+ T
 393 cell in the lung under the same conditions.

394 To conduct the computational experiment, the following conditions were taken into consideration: (i)
 395 the absence of distant tumor Ag in the lung, leading to the shutting down of the TCR signal ($U = 0$ in the
 396 equations (SI-2.1a - d), (ii) the abundance of inflammatory cytokines, including $\text{TNF}\alpha$ and $\text{IFN}\gamma$, known
 397 to induce the expression of both PD-1 and PD-L1 (SI-1), and (iii) the abundance of IL-2, which induces
 398 Blimp-1 (SI-1).

399 To account for the abundance of the lumped $\text{TNF}\alpha$ and $\text{IFN}\gamma$ species, we have replaced the rate constant
 400 σ_p in the equation (SI-2.1b) by the rate expression (SI-2.6). To account for the abundance of IL-2 in the lung
 401 compartment, we have increased the value of the parameter a_b by a factor γ in equation (SI-2.1d). In this
 402 case, we assumed that IL-2 was secreted by activated T cells (Ahmed and Nandi, 2011) and, hence, IL-2
 403 affected Blimp-1 expression through autocrine and paracrine signaling, depending on the TCR activation
 404 strength.

405 In the case when the value of the parameter γ was set to one, the level of PD-1 was increased by a factor
 406 of 6 compared with the maximum level of PD-1 shown in Fig. 7 for both anti-influenza and anti-melanoma
 407 cases. So, we can conclude that just the PD-1 DIFFL alone is not enough to counteract the effect of the
 408 pro-inflammatory cytokines. Only when a “strong action of IL-2” was taken into consideration by setting
 409 $\gamma > 5000$, the level of PD-1 was suppressed for anti-influenza T cells.

410 3 DISCUSSION

410 Below we discuss our modeling studies conducted in order to complement our immunobiochemical
 411 reconstruction towards a better understanding of the previously unrecognized acute non-oncogenic infection
 412 factor (Kohlhapp et al., 2016). We then discuss potential implications of our research to further stimulate
 413 ongoing efforts towards developing and improving physiological and functional cure approaches based on

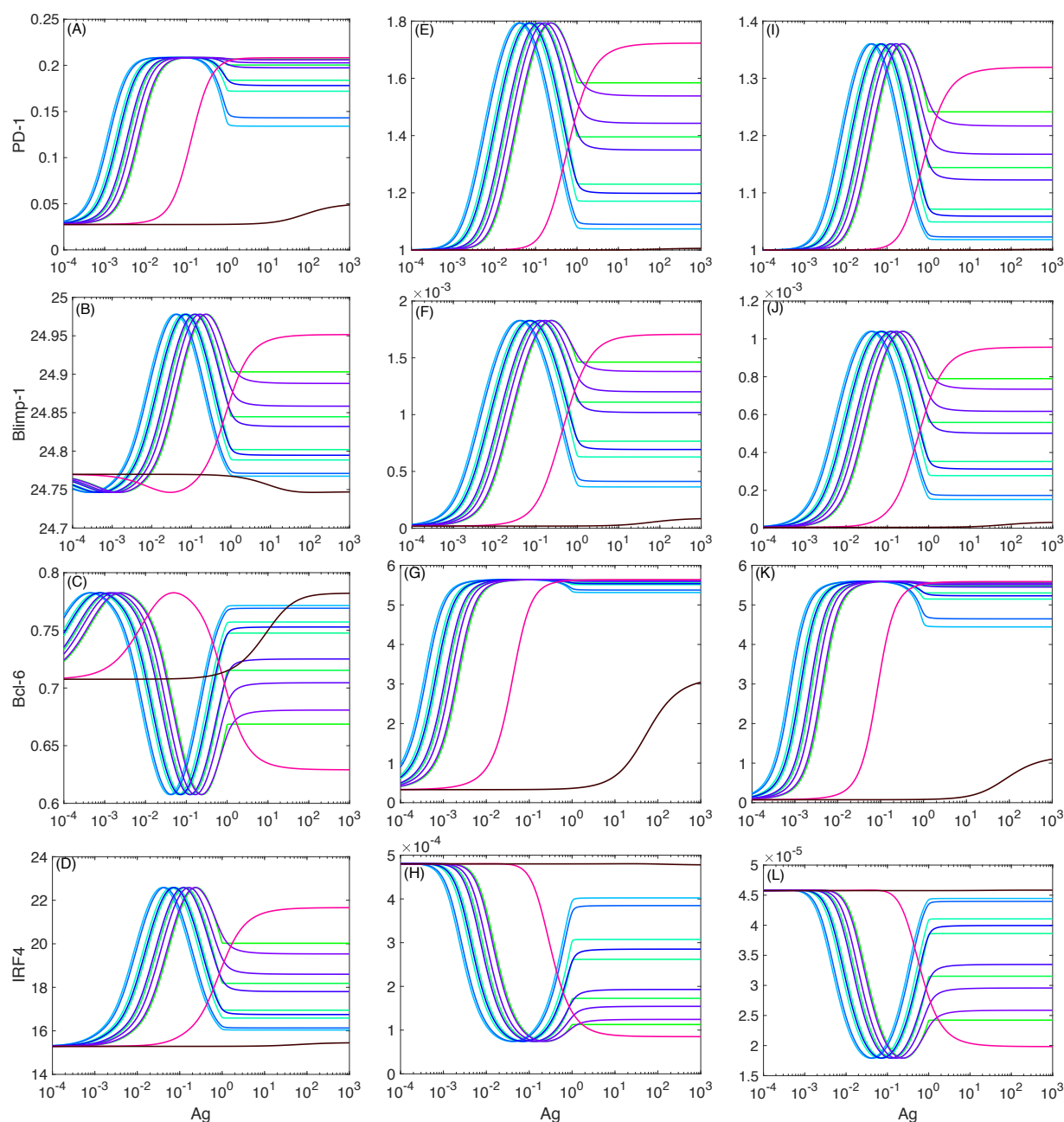


Figure 7. PD-1 DIFFL responses in the presence of PD-1:PD-L1 interaction. Panels (A) - (D) correspond the case when 20% of PD-1 receptors are ligated with PD-L1. Panels (E) - (H) correspond the case when 50% of PD-1 receptors are ligated with PD-L1. Panels (I) - (L) correspond the case when 90% of PD-1 receptors are ligated with PD-L1. All other explanations are provided in the legend for Fig. 5.

the host's ability to eliminate non-self foreign invaders and, at the same time, the host's inability to install strong altered-self (cancer) responses (Zloza, 2018).

3.1 What we learn from the model

Our PD-1 DIFFL reconstruction (Figs. 2), when combined with the mathematical modeling (Figs. 5 and 7), suggests that it is the loss of Ag dose-dependent adaptation of the expression of PD-1 receptors in the anti-tumor CD8+ T cells that could be one of major factors resulting in the multiple effects in

the presence of acute non-oncogenic infection (Kohlhapp et al., 2016). Specifically, in the case of acute infection, the level of PD-1 receptors on the surface of Ag-experienced anti-infection CD8+ T cells first increases and then decreases to lower levels in the course of the virus replication (Fig. 8(B)), the hallmark of a fundamental biological adaptation (Grossman and Paul, 1992). Therefore, based on the discussion around Fig. 3, we can conclude that chances that the cells with the phenotype shown in Fig. 8(B) will lose their motility due to PD-1:PD-L1 interactions in the infected lung are low (Fig. 4).

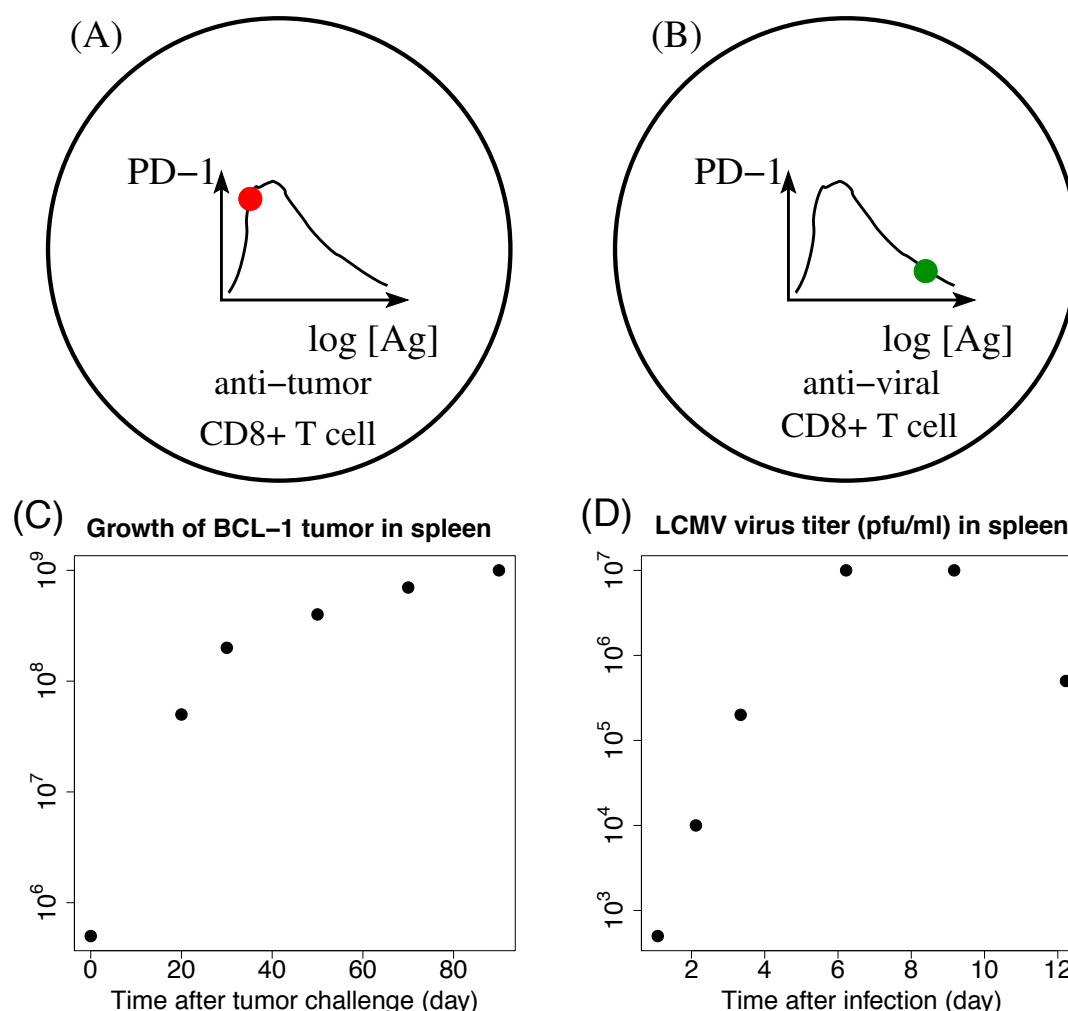


Figure 8. Schematic illustration of the adaption loss/gain hypothesis. Solid filled circles on the corresponding graphs of PD-1 receptor levels (top panels), plotted versus the log-concentrations of Ag, correspond to the levels of PD-1 receptors on anti-melanoma (A) and anti-virus (B) CD8+ T cells, respectively (top panels). Phenotype (B) corresponds to a fully developed adaptation of the PD-1 expression with respect to the increasing levels of Ag, while phenotype (A) is characterized by the lack of such adaptation. Bottom panels (C) and (D) show time-dependent levels of BCL-1 tumor cells (left) and LCMV virus titers (right) in the spleen. The data points are digitized from the corresponding plots in (Kuznetsov et al., 1994) and (Bocharov et al., 2018), respectively. Comparing panels (C) and (D), we observe that the changes in the tumor Ag levels within the first seven days are small, corresponding to the fold change less than 10 as seen from panel (C). At the same time, the viral Ag levels change significantly, corresponding to the 10^4 -fold increase during the first seven days as seen from panel (D). The small 7-day tumor Ag-level increase shown in panel (C) corresponds to the red solid “snapshot” circle in panel (A), while the large 7-day increase in the viral Ag level shown in panel (D) corresponds to the green solid “snapshot” circle in panel (B). Additional detailed explanations of panels (A) - (D) are provided in the main text.

In contrast, in the case of Ag-experienced anti-tumor CD8+ T cells, due to the much smaller levels of tumor antigens presented with MHCs in the TME, the strength of the TCR signal in anti-tumor CD8+ T cells may not be enough to activate Blimp-1 and IRF4 species to suppress PD-1 expression (Figs. 2 and 3). The lack of the expression of Blimp-1 in melanoma is known experimentally (Speiser et al., 2016). As a result, chances that T cells bearing large numbers of PD-1 receptors (Fig. 8(A)) will be paralyzed in the infected lung due to PD-1:PD-L1 interactions are high.

Importantly, the higher levels of PD-1 receptors on anti-melanoma CD8+ T cells compared with much lesser levels of PD-1 receptors on anti-influenza CD8+ T cells co-localized in the same infected lung were observed in (Kohlhapp et al., 2016). This supports the two different phenotypes shown in panels (A) and (B) of Fig. 8, respectively.

Our quantitative estimates obtained from the model (Figs. 5 and 7) show that the Ag level should be increased by several orders of magnitude required to move the Ag-experienced T cell from phenotype (A) to phenotype (B) (Fig. 8). This means that at least a 1000-fold increase in cognate Ag levels (Figs. 5 and 7) may be required for the adaptation of PD-1 expression to strong antigen-mediated stimulation.

Although more research into the novel adaptation effect illuminated by our model as well as into the lymph motion (Grebennikov and Bocharov, 2017; Tretyakova et al., 2018) and molecular mechanisms by which cells are rapidly moved with the blood (McNamara et al., 2017) is undoubtedly needed, we believe that it is worth providing some “biological” numbers that support our findings. For example, for the LCMV system, a gold standard for infectious biology, the virus titer was increased by factor about 10^3 from day 2 to day 5 (Bocharov et al., 2018, Fig.4.4). We digitized the corresponding data points and plotted them in Fig. 8 next to panel (B). Similar data are reported for influenza A infection (Bocharov and Romanyukha, 1994; Smith and Perelson, 2011). The examples of the population measurements are well translated to our modeling studies because in all cases we use dimensionless ratios of the corresponding concentrations.

Of course, one also needs to make sure whether a T cell would be capable to provide a large number of TCRs sufficient to accommodate the above huge increase in Ag-levels. Indeed, the typical number of TCR molecules is estimated in the range of 3×10^4 (Lipniacki et al., 2008), which is a reasonable number to match up with the model-suggested transition from phenotype (A) to phenotype (B) shown in Fig. 8. At the same time it is highly unlikely for tumor cells to divide as fast as the viruses do to build enough antigen that would be sufficient to change phenotype (A) to phenotype (B) within a few days. Indeed, the doubling time for virus particles can be 43 - 65 minutes (Horsfall, 1955), while the doubling time for malignant mouse melanoma B16 cells may take up to 2.8 days or longer (Bertalanffy and McAskill, 1964; Carlson, 2003).

To support the above argument, we note that (Kuznetsov et al., 1994) uses experimental data where the number of tumor cells is increased by factor about 10^2 in the time span of just 40 days. We digitized the corresponding data points and plotted them in Fig. 8 next to panel (A). We can thus conclude that due to our modeling estimations (Figs. 5 and 7), such a slow increase in Ag levels may not be enough to change between the prototypes shown in Fig. 8 for short periods of time (days), when acute infection develops and is cleared (Kohlhapp et al., 2016). Similar data can be learned from other independent studies (Radunskaya et al., 2018).

Note that the discussed transient elevation of PD-1 receptor levels as function of antigen, Fig. 8 (A, B), is experimentally observed and is also used as a “window of opportunity” in the context of the combined radiotherapy (RT) and anti-PD-1:PD-L1 treatments (Kosinsky et al., 2018). Our theoretical work provides additional valuable insight into, and add in the development of combined RT/anti-PD-1:PD-L1 therapy.

467

468 3.2 Harnessing anti-infection and anti-bacterial responses against cancer

469 By addressing the “the previously unrecognized acute non-oncogenic infection factor” revealed through
470 systematically collected heterogeneous experimental data encompassing different pathogens and tumor
471 types (Kohlhapp et al., 2016), we have suggested and discussed concrete molecular mechanisms which
472 allowed us to delineate inherently weak anti-cancer (*i.e.*, altered-self) immune responses from inherently
473 strong anti-infection (*i.e.*, non-self, foreign) responses, including co-infections.

474 Our findings may thus have potential clinical relevance particularly in the context of ever-expanding
475 immunotherapy efforts and FDA approvals involving PD-1/PD-L1 axis immune checkpoint blockade.
476 Two relevant scenarios to consider, include 1) that patients with cancer treated with such blockade may
477 also be experiencing a concomitant diagnosed or sub-clinical undiagnosed infection in a tissue distant
478 to their tumor, and 2) that selective patients with cancer are being treated with oncolytic viruses (OVs),
479 which preferentially infect tumor cells, but can also infect cells in tissues distant to their tumor (Kaufman
480 et al., 2015; Jhavar et al., 2017). In both scenarios, checkpoint blockade may have less-recognized effects
481 discussed here (e.g., releasing the T cell motility paralysis caused by an infection in a tissue distant to the
482 tumor) and thus such blockade may improve patient outcomes, including in the context of combination
483 with OV (Samson et al., 2018). As additional clinical information is collected from patients receiving
484 checkpoint blockade (including about infection status and OV viral loads in non-injected sites), future
485 efforts may provide the data necessary to reveal and model this blockade effect further.

486 We conclude this work with a hope that our theoretic analysis of the newly discovered infection-tumor
487 interaction (Kohlhapp et al., 2016), made by combining solid immunobiochemical reconstruction with
488 appropriate mathematical modeling may also be useful in current developments of both “physiological”
489 and “functional cures” (Zloza, 2018). Specifically, our mechanistic molecular-based analysis of the novel
490 immunologic phenomenon uncovers important competing push-pull processes fundamentally inherent
491 in immunology (Grossman and Paul, 1992, 2001, 2015). We believe that the results reported may have
492 broader implication towards developing (*i*) *physiological cure* approaches in order to completely eliminate
493 tumors as it happens in the case of rapid (one week long) clearance of acute infection, and, alternatively,
494 towards undertaking (*ii*) *functional cure* treatments to maintain long-term immunologic control as in the
495 cases of controlled chronic infection and other disorders as, for example, hypertension (Dampney et al.,
496 2007). However, research (Kohlhapp et al., 2016) clearly suggests that all such cures must be developed
497 with care.

CONFLICT OF INTEREST STATEMENT

498 The authors declare that the research was conducted in the absence of any commercial or financial
499 relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

500 A.Z. and E.D.S. conceived the work. A.Z. contributed data. E.V.N. and E.D.S. designed research. E.V.N.
501 analyzed the data and performed the research. E.V.N. and E.D.S. wrote the manuscript.

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SUPPLEMENTAL DATA

503 An extended literature review of the corresponding mechanistic detail, the model construction, and the
 504 model's parameter justification can be found in Supplementary Information (SI).

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Supplementary Material

Immunobiochemical Reconstruction of Influenza Lung Infection - Melanoma Skin Cancer Interactions

SI-1 A REVIEW OF RELEVANT LITERATURE

SI-1.1 An immunobiochemical reconstruction scope

Given the complexity of the processes relevant to the observations (O1) - (O5), we must limit the scope of our research. Because the interaction between PD-1 and PD-L1 is regarded as a major “T cell brake” (Nirschl and Drake, 2013), and because it is also a central topic in (Kohlhapp et al., 2016), we center our analysis on the construction of the gene regulatory networks (GRNs) directly involved in the expression of PD-1 receptors under different immunologic contexts.

A very important topic that shapes the scope of our work is that T cell activities are controlled at multiple levels. These regulatory controls are necessary to prevent T cells from becoming hyperactivated, causing significant collateral damage to non-target tissue. These types of responses enhance inflammation, resulting in the release of self-antigens from necrotic tissue, increasing the chances for the induction of autoimmune diseases (Liechtenstein et al., 2012). To avoid autoimmunity induced by necrotic tissue, key regulatory T cell inhibitory interactions occur between PD-L1 expressed on immune, infected and tumor cells (Nirschl and Drake, 2013), and, PD-1 expressed on T cells (Sakaguchi et al., 2008; Fife et al., 2009; Francisco et al., 2010; Nirschl and Drake, 2013; Schietinger and Greenberg, 2014; Bardhan et al., 2016; Sharpe and Pauken, 2017).

SI-1.2 A TCR activation primer

For a strong CD8+ T cell activation, three well-known signals have to be provided from professional antigen presenting cells (APCs) (Kindt et al., 2007):

- Signal 1: Antigen presentation to T cells.
- Signal 2: Co-stimulation.
- Signal 3: Cytokine priming.

Signal 1 is mediated by binding of a T cell receptor (TCR) on T cells with its cognate antigen presented on an MHC.

Signal 2 is mediated by a series of receptor:ligand bindings, such as CD80 binding to CD28 between the APC and the T cell, respectively. The combination of TCR engagement, CD28 binding, and IL-2 activates Zap-70, lck and PI3K, which in turn lead to T cell activation, expansion, and acquisition of effector activities (Liechtenstein et al., 2012; Rendall and Sontag, 2017). Furthermore, in reality, as Liechtenstein et al. (2012) states, a variety of ligand-receptor interactions take place in the immunological synapse, many of which are inhibitory. The final integration between activatory and inhibitory interactions determine the type and strength of the co-stimulatory signal given to the T cells, setting the “degree” of T cell activation.

Signal 3 is mediated by binding of cytokines to their respective receptors, such as IL-2 produced by T cells binding to IL-2 receptors also on the same T cells.

During antigen presentation to naïve T cells, PD-1:PD-L1 interaction acts as a brake in TCR signal transduction (Nirschl and Drake, 2013). PD-1 is transiently up-regulated during antigen presentation as a consequence of T cell activation (Freeman et al., 2000) and PD-1:PD-L1 binding results in ligand-induced TCR down-modulation (Escors et al., 2011; Karwacz et al., 2011, 2012).

To this end, Liechtenstein et al. (2012) suggest that TCR down-modulation is absolutely required for T cell activation in order to prevent T cell hyperactivation by terminating TCR signal transduction. In such cases, PD-1 associates to the TCR at the immunological synapse and controls its signal transduction as well

as its presence on the T cell surface (Karwacz et al., 2011). TCR down-modulation is largely reduced when PD-L1 is silenced in antigen-presenting DCs, or when PD-1:PD-L1 is blocked using antibodies during antigen presentation (Liechtenstein et al., 2012).

Finally, as further reviewed in (Liechtenstein et al., 2012), PD-1:PD-L1 interactions control the timing of TCR stimulation in at least two different ways: (i) by removing TCRs from the T cell surface, and (ii) by terminating the intracellular signal transduction pathways after recruiting phosphatases SHP1 and SHP2 (Zhang and Rundell, 2006; Bardhan et al., 2016). Note briefly that processes in (ii), put, for example, a brake on NF- κ B signaling, the inhibitory process that shuts down IRF4, which in turn removes the Blimp-1 imposed brake from PD-1 transcription, where both IRF4 and Blimp-1 are key molecular species of our analysis, as discussed below in detail.

SI-1.3 Linking observations with immunological mechanisms

(O1) Anti-tumor CD8+ T cells are shunted to the lung during influenza infection.

Here, we discuss six mechanisms (O1-M1) - (O1-M6). In order to get insight into mechanism (O1-M1), we subdivide mechanism (O1-M1-B) into two complementary immunological mechanisms (O1-M1-A) and (O1-M1-B).

Mechanism (O1-M1-A) Cytotoxic CD8+ T cells (T_{EFF}) are highly dynamic within dense tissue once activated, immediately followed by their migration arrest, induced by integrin upregulation that stops motion and promotes effective synapse formation (Feinerman et al., 2008) when contacting a high potency antigen available at adequate density on tissue resident antigen-presenting cells (APCs) (Marelli-Berg et al., 2010; Honda et al., 2014). Indeed, T cells in tissues migrate along chemotactic gradients until they encounter antigen on an APC, which leads to their Intercellular Adhesion Molecule (ICAM-1)-dependent arrest by TCR-mediated “stop signals” (Jennrich et al., 2012).

Specifically, activated T cells have been shown to rapidly traffic in tissue in response to chemokines and cytokines (Kindt et al., 2007; Chimen et al., 2017), including CXCL9, CXCL10, INF γ , etc. (Ogawa et al., 2002; Baaten et al., 2013; Oelkrug and Ramage, 2014; Kim and Chen, 2016; Spranger, 2016; Stein et al., 2016) with a primary function to find and kill target cells expressing cognate antigen (Ag) (Bhat et al., 2014).

Dynamic speed and travel patterns of T_{EFF} are predominantly influenced by the tissue environment rather than by mechanisms intrinsic to the T_{EFF} . Specifically, activated T cells have been shown *in vivo* to traffic to inflamed skin, even in the absence of cognate Ag (Biotec and Gladbach, 2011), suggesting that Ag alone does not play an essential role in the recruitment of circulating CD8+ T cells (Van Braeckel-Budimir and Harty, 2017).

To this end, a natural question arises: “Why is it the anti-tumor T_{EFF} cells that are shunted to the infected lung, and not vice versa, that is, why is it not the anti-influenza T_{EFF} cells that are shunted to the tumor compartment instead?”

Mechanism (O1-M1-B) The above question can be addressed by reviewing *in vivo* studies describing T cell Ag-induced arrest in tissues in direct proportion to the amount of Ag present (Beattie et al., 2010; Deguine et al., 2010; Celli et al., 2011; Honda et al., 2014). Indeed, the time needed for the T_{EFF} killing of highly antigenic cells *in vivo* through the cell-to-cell attachment and TCR-pMHC (Ag) binding events can be partially attributed to effective half-life or “confinement time” of a TCR-pMHC interaction (Aleksic et al., 2010).

The process can last for long periods of time depending on the context, (i) in the range of 40 minutes, (ii) over 3-6 hours and (iii) up to 48 hours, in order to form “stable immunologic synapses” (Grakoui et al., 1999; Liechtenstein et al., 2012; Xie et al., 2013; Tkach and Altan-Bonnet, 2013; Ortega-Carrion and Vicente-Manzanares, 2016; Stein et al., 2016), which are needed to complete a series of signaling events, including co-receptor recruitment and TCR phosphorylation (McKeithan, 1995; Garcia et al., 2007; Breart et al., 2008; Tkach and Altan-Bonnet, 2013; Liu et al., 2014; Das et al., 2015; Parello and Huseby, 2015).

Although T cells become rapidly arrested after their contact with Ag-presenting tissue-resident cells, their subsequent recovery period was found to be heterogeneous, with some cells regaining motility within 30 min and others remaining arrested for several hours (Honda et al., 2014). After the extended arrest, T cells can be oscillating between periods of brief arrest and motility, suggestive of additional TCR stimulation, before regaining a migration pattern similar to what was observed in the absence of antigen (Honda et al., 2014).

Such Ag-induced arrested T cells are functionally very distinct of highly motile T cells, because the arrested T cells were found to be characterized by a profoundly increased production of $\text{INF}\gamma$ (Honda et al., 2014).

All this suggests that T cell Ag-induced arrest on target cells positively correlates with their effector function, and the balance between motility and Ag-induced arrest controls T cell activation (Stein et al., 2016).

Here, the definitions of “long periods of time” and “stable immunologic synapses” should be understood dynamically and not statically in terms of “fast association and dissociation rates” (Coombs et al., 2011; Tkach and Altan-Bonnet, 2013) that define the “confinement time” of a TCR-pMHC interaction (Aleksic et al., 2010). Multiple studies have indicated that T cells integrate these discontinuous antigen contacts over time and respond in proportion to the cumulative duration of TCR signaling as reviewed in (Tkach and Altan-Bonnet, 2013).

Many tumor-specific antigens provoke only weak immune responses, which are incapable of eliminating all tumor cells (Aleksic et al., 2010). This is in line with the McKeithan-Altan-Bonnet-Germain kinetic proofreading model (KPL-IFFL) (Hopfield, 1974; McKeithan, 1995; Altan-Bonnet and Germain, 2005; François et al., 2013; Courtney et al., 2017), which is based on the well-established fact that T cell activation is selected by evolution to discriminate a few foreign peptides rapidly from a vast excess of self-peptides (François et al., 2013). We use the abbreviation KPL-IFFL for the kinetic proofreading model coupled with limited signaling and incoherent feedforward loop (Lever et al., 2016).

Because many tumor antigens are self antigens (Liechtenstein et al., 2012), often called Tumour-Associated Antigens (TAAs) and Tumor-Specific Antigens (TSA) (Kindt et al., 2007), anti-tumor TCRs may be of lower affinity due to their selection and training against “self”-antigen reactivity (Hogquist and Jameson, 2014) compared with those TCRs evolved to recognize viral epitopes (Irving et al., 2012; Vonderheide and June, 2014). Indeed, the affinity of TCR clones for novel or not previously encountered antigens, like tumor antigens, is remarkably low, typically 1 - 10 μM (Courtney et al., 2017). This is in line with a commonly accepted fact that high-affinity tumour-specific TCR clonotypes are typically deleted from the available repertoire during thymic selection because the vast majority of targeted epitopes are derived from autologous proteins (Tan et al., 2015).

This phenomenon is known as “antigen discrimination” (Galvez et al., 2016; Rendall and Sontag, 2017), and is currently discussed in terms of the “antigen-receptor (catch bonds) lifetime dogma” (Feinerman et al., 2008; François and Altan-Bonnet, 2016).

In addition to the TCR antigen discrimination, tumors do not express large amounts of cognate Ag on MHCs to keep systemic tolerance and prevent the development of autoimmune diseases (Liechtenstein et al., 2012; Nirschl and Drake, 2013), compared with viral infection.

Antigen expression by tumor cells thus determines T_{EFF} motility within the tumor (Boissonnas et al., 2007). Such mobile T_{EFF} cells can follow collagen fibers or blood vessels, or migrate along blood vessels preferentially adopting an elongated morphology (Boissonnas et al., 2007), when nothing can prevent them from freely moving along high infection-induced chemokine gradients toward the infected and inflamed lung.

We finalize the description of this mechanism by pointing out a very important and relevant study (Poleszczuk et al., 2016) which documents intensive motility of anti-tumor T_{EFF} cells when the anti-tumor T_{EFF} cells enter and leave the TME back to the bloodstream and lymph multiple times before they get finally arrested and absorbed by the multiple-time revisited TME.

Mechanism (O1-M2) Tumors themselves may induce emigration of tumor-specific CD8+ T cells from the TME by employing chemokines like SDF-1/CXCL12 (Marelli-Berg et al., 2010; Joyce and Fearon, 2015; Kim and Chen, 2016). When the level of CXCL12 becomes greater than the levels of other chemoattractors, CXCL12 acts as chemorepeller (Marelli-Berg et al., 2010; Vianello et al., 2005), that is, at low levels, CXCL12 is a chemoattractor, while at high levels CXCL12 is a chemorepeller. Expression of CXCL12 and its receptor CXCR4 is induced by $IFN\gamma$ (Ogawa et al., 2002; de Oliveira et al., 2013).

Sometimes, this effect is called *fugetaxis* (Vianello et al., 2005). Because $IFN\gamma$ is produced in large quantities by different cell types in inflamed infected sites (Kindt et al., 2007), and then circulates to the tumor site within the bloodstream, it can be concluded that (distant) infection can significantly contribute to the egress of anti-tumor CD8+ T cells from the TME.

Mechanism (O1-M3) Due to non-specific cardiovascular edema effects, caused by infection-induced inflammation (Marchuk, 1997), the general circulation pattern of central memory (T_{CM}) and naïve T cells (Donnadieu, 2016; Levin et al., 2016) throughout the body from blood, across high endothelial venules (HEVs) into lymph nodes, through T cell zones, out via efferent lymphatics, and eventually back into the blood through the thoracic duct is significantly perturbed and is redirected to the site of infection-induced inflammation (Marchuk, 1997; Levin et al., 2016).

Mechanism (O1-M4) As mentioned earlier, different cells in infected tissues induce cytokine production (Kindt et al., 2007). Cytokines play multiple roles such as chemoattraction of dendritic cells, macrophages, T cells, NK cells, and promotion of T cell adhesion to endothelial cells (Dufour et al., 2002). To this end, significant levels of both activated influenza-specific and non-specific T cells were found present in infected lung and measured (Toapanta and Ross, 2009).

The inflammatory chemokine receptor CXCR3 has been recently identified with effective T cell function. CXCR3 expression is increased during T cell activation and is important for homing to inflammatory sites. Its ligands CXCL9, CXCL10 and CXCL11 are rapidly induced during inflammation and guide T cells into specific microenvironments in lymphoid and non-lymphoid tissues as reviewed in (Stein et al., 2016).

Mechanism (O1-M5) High levels of IL-2 produced by activated anti-infection CD8+ T cells at the infection site counteract the repelling action of CXCL12 (Beider et al., 2003) in contrast to the opposite effect elicited by CXCL12 in the TME as discussed earlier.

Mechanism (O1-M6) The PD-1 mediated control of immune responses depends on interactions between PD-1 on CD8+ T cells and PD-L1 in tissues (Nirschl and Drake, 2013), inducing CD8+ T cell motility

paralysis via PD-1:PD-L1 stable bonds (Zinselmeyer et al., 2013; Schietinger and Greenberg, 2014; Stein et al., 2016). We introduce the “paralysis” mechanism (O1-M6) into the context of our studies by discussing systematically the following specific questions,

- (Q.1) “What triggers expression of PD-1 receptors on CD8+ T cells, and why is the expression triggered in the first place?”
- (Q.2) “Why are PD-1 receptors over-expressed in larger quantities on anti-tumor CD8+ T cells and not on anti-influenza CD8+ T cells co-localized within the same infected lung?”
- (Q.3) “Why are anti-VACV CD8+ T cells not sequestered in the infected lung when the host is distantly co-infected with both infections, influenza A and VACV infections (in the absence of tumors)?”

To address (Q.1), we follow Simon and Labarriere (2017) who reviewed results highlighting the ambiguous role of PD-1 in defining efficient or inefficient adaptive immune response. Initially, PD-1 transient expression on native T cells is induced immediately upon TCR activation, that is, the number of PD-1 receptors can be regarded as a biomarker of activated and *not* exhausted CD8+ T cells.

The level of PD-1 receptors decreases in the absence of TCR signaling but is maintained upon chronic activation with a persistent epitope target such as in chronic viral infections and in cancer (Wherry et al., 2007; Brown et al., 2010; Pauken and Wherry, 2015b;a). Thus, the number of PD-1 receptors can *also* be regarded as a biomarker of exhausted T cells (Simon and Labarriere, 2017). Importantly, transient expressions of PD-1 and PD-L1 is viewed as a window of opportunity in the combined radiation (RT) and anti-PD-1:PD-L1 therapies (Kosinsky et al., 2018).

The discussed ambiguous role of PD-1, which can be viewed either as a biomarker of activated or exhausted CD8+ T cells depending on the inflammation context, can be explained as follows.

First, although the central immune tolerance mechanism results in the removal of most of the auto- or self-reactive T cells during thymic selection, a fraction of self-reactive lymphocytes escapes to the periphery and poses the threat of autoimmunity. Moreover, “it is now understood that the T cell repertoire is in fact broadly self-reactive, even self-centered” (Hogquist and Jameson, 2014; Grossman and Paul, 2015; Richards et al., 2016). The strength with which a T cell reacts to self ligands and the environmental context in which this reaction occurs influence almost every aspect of T cell biology, from development to differentiation to effector function (Hogquist and Jameson, 2014; Grossman and Paul, 2015).

The immune system has evolved various mechanisms to constrain autoreactive T cells and maintain peripheral tolerance (Grossman and Paul, 2001, 2015), including the constitutive expression of PD-L1 in large quantities in various tissues (*e.g.*, lungs, pancreatic islets, *etc.*), and T cell anergy, deletion, and suppression by regulatory T cells (Sakaguchi et al., 2008; Fife et al., 2009; Francisco et al., 2010; Schietinger and Greenberg, 2014; Bardhan et al., 2016).

Second, although T cells endow their host with a defense that favors pathogen clearance, this efficiency sometimes gives rise to intolerable immunopathology, especially when a pathogen transitions into a state of persistence. For this reason, the immune system is equipped with dampening mechanisms that induce T cell exhaustion via PD-1 and PD-L1 immune regulators (Zinselmeyer et al., 2013; Pauken and Wherry, 2015b; Bardhan et al., 2016). This means that the activated T cells must be attenuated irrespective of whether invaders are eliminated or persist. This is because, quite often, persisting microorganisms may cause less tissue damage than the associated immunopathology as a result of continued lymphocyte cytotoxicity (Speiser et al., 2016).

Overall, this means that the immune system prefers to put infection that cannot be eradicated rapidly into a chronic state which should produce less damage to the body than the extended exposition of the body to aggressive CD8+ T cell response (Grossman and Paul, 1992, 2015).

(O2) Disruption of anti-tumor responses is not due to tumor-induced immune suppression of viral clearance or the inability of the immune system to respond to concomitant challenges.

The observation (Kohlhapp et al., 2016) that cancer does not significantly suppress the natural anti-viral response can be explained by similar arguments used to introduce the mechanisms (O1-M1) - (O1-M6). These suggest that much weaker inflammation in the tumor site compared with much stronger inflammation in the infected lung may not be enough to force the anti-influenza CD8+ T cells (arrested in the lung as discussed earlier) to egress the lung. The observation (Kohlhapp et al., 2016) that influenza infection did not alter the natural clearance of the VACV or the proportion of VACV-tetramer+ CD8+ T cells at the site of influenza infection can also be explained by the local anti-VACV CD8+ T cells Ag-induced arrest required to kill VACV-infected cells as discussed earlier.

(O3) Therapeutic blockade of PD-1 results in reversal of infection-mediated anti-tumor response disruption.

Recall that PD-L1 promotes motility paralysis (Zinselmeyer et al., 2013; Schietinger and Greenberg, 2014; Stein et al., 2016). In other words, the bond PD-1:PD-L1 mediates locking T cells into a state of prolonged motility paralysis by localizing to the environment with abundant PD-L1 expression on stromal cells as discussed earlier, termed “T cell motility paralysis” in (Zinselmeyer et al., 2013; Schietinger and Greenberg, 2014).

Because the bond PD-1:PD-L1 is formed dynamically due to interchanging binding and unbinding processes, blockade of PD-1 shifts the dynamic equilibrium towards dissociation of PD-1:PD-L1 bond, leading to the rapid recovering (about 30 min.) of T cell motility, signaling, and cytokine production (Zinselmeyer et al., 2013; Oelkrug and Ramage, 2014; Pauken et al., 2016). The corresponding details are summarized in Table SI-1.1 of Sect. SI-1.4.

Reactivated anti-tumor CD8+ T cells then detach from local PD-L1 anchors and start moving with lymph outside of the infected lung and may ultimately return back to the tumor site (Calzascia et al., 2005) with the blood flow, following similar trafficking routes and mechanisms as discussed in (Poleszczuk et al., 2016).

SI-1.4 Reactivation of exhausted effector cells

Because the PD-1 blockade reactivates exhausted anti-tumor CD8+ T cells, sequestered in the infected lung and return, possibly, them back to the TME (Kohlhapp et al., 2016), we briefly summarize relevant known results on the exhausted T cell reactivation (Table SI-1.1). Our summary is based on the recent reports (Zinselmeyer et al., 2013; Pauken et al., 2016; Wang et al., 2017).

Several gene signatures based on the analyses of populations of dysfunctional CD8+ T cells from cancer and chronic viral infections have been published and reviewed by Wang et al. (2017). These signatures confirm great similarity between virus- and cancer-associated CD8+ T cell dysfunction. Due to these published gene signature comparisons, we believe that Table SI-1.1 further supports mechanisms formulated in the main text.

Table SI-1.1. A brief summary of T_{EX} reactivation after PD-1 blockade.

Reactivation Effect	Mechanisms
(E.1) reactivation of T_{EFF} functions in T_{EX}:	
(E.1.1) <i>improved cell cycle and proliferation</i>	• increased transcription of cell division genes • increased levels of Ki-67
(E.1.2) <i>improved response to antigen</i>	• elevated co-production of INFγ and TNFα
(E.1.3) <i>improved motility and chemotaxis</i>	• upregulated expression of <i>cxc19</i> and <i>cxc3</i>
(E.1.4) <i>improved killing capability</i>	• increased levels of granulocytes
(E.1.5) <i>suppression of PD-1 expression</i>	• upregulated expression of <i>prdm1</i> encoding Blimp-1
(E.1.6) <i>protection against exhaustion</i>	• upregulated expression of <i>il7r</i> • OCR state specific regulation of <i>ctla4</i>
(E.2) reactivation of T_{MEM} functions in T_{EX}:	
(E.2.1) <i>negative regulation of apoptosis</i>	• increased levels of phospho-STAT5
(E.2.2) <i>improved adhesion</i>	• unknown
(E.2.3) <i>improved regulation of activation</i>	• elevated production of INFγ
(E.3) transient reinvigoration of T_{EX} (peaked in 3-weeks)	
<i>transient and Ag-dose dependent expression of prdm1 encoding Blimp-1</i>	• small and large amounts of Ag repress <i>prdm1</i> • medium amounts of Ag activate <i>prdm1</i>
(E.4) signaling and immunometabolic effects:	
(E.4.1) <i>signaling</i>	• upregulation of genes encoding NF-κB and IRFs
(E.4.2) <i>lipid metabolism</i>	• upregulation of genes encoding PPARγ and RXRα • downregulation of <i>srebp1</i>
(E.4.3) <i>de-novo cholesterol pathway, and glycolysis</i>	• unknown

SI-2 A CORE MATHEMATICAL MODEL OF PD-1 EXPRESSION

SI-2.1 The model equations

Our core mathematical model of PD-1 expression on the surface of a CD8+ T cell describes normal and aberrant dynamics of interactions between four immunobiochemical entities, Bcl-6 (C), PD-1 (P), IRF4 (I), and Blimp-1 (B),

$$\underbrace{\frac{dC}{dt}}_{C=[\text{Bcl6}]} = \underbrace{\left(\frac{a_c U^{n_c}}{A_c^{n_c} + U^{n_c}} \right)}_{\text{TCR dep. act.}} \underbrace{\left(\frac{M_c^{r_c}}{M_c^{r_c} + B^{r_c} + I^{r_c} + C^{r_c}} \right)}_{\text{Blimp1/IRF4/Bcl6 dep. repr.}} - \underbrace{\mu_c C}_{\text{Bcl6 deg.}} \quad (\text{SI-2.1a})$$

$$\underbrace{\frac{dP}{dt}}_{P=[\text{PD1}]} = \left(\sigma_p + \underbrace{\frac{a_p U^{n_p}}{A_p^{n_p} + U^{n_p}}}_{\text{TCR dep. act.}} \right) \underbrace{\left(\frac{M_p^{r_p}}{M_p^{r_p} + B^{r_p}} \right)}_{\text{Blimp1 dep. repr.}} - \underbrace{\mu_p P}_{\text{PD1 deg.}} \quad (\text{SI-2.1b})$$

$$\underbrace{\frac{dI}{dt}}_{I=[\text{IRF4}]} = \left(\sigma_i + \underbrace{\frac{a_i U^{n_i}}{A_i^{n_i} + U^{n_i}}}_{\text{TCR dep. act.}} + \underbrace{\frac{k_i B^{m_i}}{K_i^{m_i} + B^{m_i}}}_{\text{Blimp1 dep. act.}} + \underbrace{\frac{q_i I^{s_i}}{Q_i^{s_i} + I^{s_i}}}_{\text{IRF4 dep. act.}} \right) \Phi_L - \underbrace{\mu_i I}_{\text{IRF4 deg.}} \quad (\text{SI-2.1c})$$

$$\underbrace{\frac{dB}{dt}}_{B=[\text{Blimp1}]} = \left(\underbrace{\frac{a_b U^{n_b}}{A_b^{n_b} + U^{n_b}}}_{\text{TCR dep. act.}} + \underbrace{\frac{k_b I^{m_b}}{K_b^{m_b} + I^{m_b}}}_{\text{IRF4 dep. act.}} \right) \underbrace{\left(\frac{M_b^{r_b}}{M_b^{r_b} + C^{r_b}} \right)}_{\text{Bcl6 dep. repr.}} - \underbrace{\mu_b B}_{\text{Blimp1 deg.}} \quad (\text{SI-2.1d})$$

Here, for the sake of compactness in the equation term explanation, we use the following abbreviations, “TCR dep. act.” for TCR-dependent activation, “Blimp-1/IRF4/Bcl-6” for Blimp-1/IRF4/Bcl-6-dependent repression, and so on.

The model structure corresponds to the circuit topology depicted in Fig. 2 of the main text with a few simplifications resulting from lumping some species, (i) NFATc1 and PD-1 becoming the species P , and (ii) NF- κ B and IRF4 becoming the species I . We also omit Erk-dependent degradation of Bcl-6 because it is in turn attenuated by Bcl-6 itself.

The input $U := U(\alpha, \kappa, P)$ to the model (SI-2.1) is described by the scalar function $u(\alpha, \kappa)$ defined in (SI-3.11),

$$U(\alpha, \kappa, P) = u(\alpha, \kappa) \phi_L(P), \quad (\text{SI-2.2a})$$

$$\phi_L(P) = \frac{H_p}{H_p + L P}. \quad (\text{SI-2.2b})$$

Here, the inhibitory regulatory factor $\phi_L(P)$ corresponds to the co-localization of PD-1:PD-L1 complexes around the immunologic Ag-TCR synapses that hinder the TCR activity as discussed in Sec. SI-1.2. An external environment parameter L models a fraction of PD-1 receptors bound with PD-L1. Parameters α and κ are scaled Ag level and scaled k_{off} , the dissociation constant for the Ag-TLR bond, respectively.

We use a Michaelis-Menton saturation functional dependence in the expression (SI-2.2b) to describe a 2D-sliding diffusion of PD-1 receptors on the surface of a T cell (without any switch-like reaction sharp transitions) as a major process contributing to the TCR down regulation effect (Sect. SI-1.2).

Next, the factor $\Phi_L := \Phi_L(P)$ in the equation (SI-2.1c) describes a net negative feedback effect caused by the PD-1:PD-L1 interaction (Sec. SI-1.2),

$$\Phi_L(P) = \frac{H_L^{h_L}}{H_L^{h_L} + (LP)^{h_L}}. \quad (\text{SI-2.3})$$

Recall that the active complex formed between PD-1 and PD-L1 suppresses the NF- κ B pathway, while the NF- κ B pathway activates IRF4 (Fig. 2).

We also use generic Hill functions in (SI-2.2b) and (SI-2.3) following the Hill-function approximation suggested for T cell exhaustion in (Johnson et al., 2011).

In order to capture effects caused by self (tumor) and non-self (infection) interactions, including significant differences in the magnitude of infection and amount of tumor antigens, we implement the following relationships to mathematically implement the self- / non-self specificity,

$$\alpha_T < \alpha_I, \quad (\text{SI-2.4a})$$

$$\kappa_T > \kappa_I. \quad (\text{SI-2.4b})$$

Here, subscript labels “T” and “I” correspond to tumor and infection, respectively.

Based on our immunobiochemical reconstruction and following (Warmflash and Dinner, 2009), we make explicitly additional choices to rank TCR-mediated activation parameters as follows,

$$A_c \leq A_p \leq A_i \leq A_b, \quad (\text{SI-2.5a})$$

$$a_c \leq a_p \leq a_i \leq a_b. \quad (\text{SI-2.5b})$$

Based on the developed immunobiochemical reconstruction, the inequality choices (SI-2.5a) ensure that the genes encoding Bcl-6 and PD-1 are activated at lower antigen levels than the genes encoding IRF4 and Blimp-1, whereas the latter ensures that the switch towards the suppression of PD-1 transcription is biased towards the CD8+ T cell, when both IRF4 and Blimp-1 are expressed at high antigen levels.

To account for the abundance of the lumped $\text{TNF}\alpha/\text{IFN}\gamma$ species, we have replaced the rate constant σ_p in the equation (SI-2.1b) by the reaction rate expression,

$$\tilde{\sigma}_p = \sigma_p + \frac{k_T T^{n_T}}{K_T^{n_T} + T^{n_T}}. \quad (\text{SI-2.6})$$

Here, T corresponds to $\text{TNF}\alpha$, $k_T = 0.5$, $k_T = 1$, and $n_T = 2$. The values of the new parameters are selected in the range of the corresponding parameter values from Table SI-2.1.

SI-2.2 The model parameters

Reference parameter values used in our modeling studies are listed in Table SI-2.1. We have to mention explicitly that the parameter values have not been fitted to any data from (Kohlhapp et al., 2016), and have been selected as follows.

Table SI-2.1. PD-1 expression model parameter values.

Parameters	Values	Comments
σ_i	0.30	IRF4 constituent synthesis rate
σ_p	0.10	PD-1 immune central tolerance const. synth. rate
$a_c = a_p < a_i < a_b$	$0.75 = 0.75 < 75.0 < 100.00$	genetic switch thresholds
$A_c < A_p < A_i < A_b$	$0.01 < 0.10 < 1.00 < 10.00$	genetic switch thresholds
$n_c = n_p$	3	species: Bcl-6 and PD-1
$n_b = n_i$	2	species: Blimp-1 and IRF4
k_b	0 – 25	species: IRF4
$k_i = q_i$	7.50	species: Blimp-1 and IRF4
$K_b = K_i = Q_i$	1.00	species: Blimp-1 and IRF4
$m_b = m_i = s_i$	2	species: Blimp-1 and IRF4
$M_b = M_c = M_p$	10.00	species: Blimp-1, Bcl-6 and PD-1
$H_p = H_L$	0.1	species: PD-1 and PD-L1
$r_b = r_c$	2	species: Blimp-1 and Bcl-6
$r_p = h_L$	4	species: PD-1 (p) and PD-L1 (L)
$\mu_c = \mu_p$	0.10	species: Bcl-6 and PD-1
$\mu_b = \mu_i$	1.00	species: Blimp-1 and IRF4
L	0 - 1	species: fraction of PD-1 bound to PD-L1

First, we used dimensionless (scaled) parameter values of the same order of magnitude for the corresponding subsets of parameters as those which were used in (Sciammas et al., 2011; Martinez et al., 2012; Lever et al., 2016).

In our selection of the reference parameter values (Table SI-2.1), we also analyzed and followed a number of insightful discussions of a very challenging and complex problem of selecting relevant parameter values for biological and especially immunological models, presented in a number of published works (Heinrich and Rapoport, 2005; Warmflash and Dinner, 2009; Martinez et al., 2012; Lever et al., 2014; Galvez et al., 2016), including conceptual views (Gunawardena, 2014; Eftimie et al., 2016) as well as discussed general issues with experimental measurements (De Boer and Perelson, 2013; Eftimie et al., 2016).

Second, the parameter values used from (Sciammas et al., 2011; Lever et al., 2016) can be justified for our modeling studies by employing the following IFFL function argument. Indeed, the incoherent feedforward loops cannot exert their biphasic function with any arbitrary parameter values (Kim et al., 2008). The parameter values taken from (Sciammas et al., 2011; Lever et al., 2016) and used in the model (SI-2.1) correspond to the dose-dependent biphasic behaviors as defined and studied in (Kim et al., 2008), and also observed experimentally in the cited literature. In other words, the used parameter values are sufficient to instill the IFFL function.

Finally, the type of modeling carried out in our work can be characterized as phenotypic modeling (Warmflash and Dinner, 2009; Lever et al., 2014; Gunawardena, 2014). Recall that the objective of the phenotypic modeling is to capture the function of a biological system, based on the available and well-established features of the regulatory network under study as also explicitly stated in (Sciammas et al., 2011) which justified the selection of generic Hill functions in their model tailored to the GRN topology. In this work, we implemented a similar approach.

SI-3 THE KPL-IFFL MODEL

For the sake of consistency in the integration of the model (Lever et al., 2016) with our model describing the core circuit, we briefly derive functional relationships needed for the models' integration, adapting the discussion in (Lever et al., 2016).

Specifically, our objective here will be to derive the function $u(\alpha, \kappa)$, which we define as a non-dimensionalized input P in (SI-3.10b), and for which the final expression is given in (SI-3.11). The scaled function $u(\alpha, \kappa)$ depends on two state variables α and κ , the scaled level of Ag and the scaled value of the off-rate constant k_{off} , respectively.

A mathematical model (Lever et al., 2016) is

$$\frac{dL}{dt} = -k_{\text{on}}LR + k_{\text{off}}C_T, \quad (\text{SI-3.1a})$$

$$\frac{dR}{dt} = -k_{\text{on}}LR + k_{\text{off}}C_T, \quad (\text{SI-3.1b})$$

$$\frac{dC_0}{dt} = k_{\text{on}}LR - (k_{\text{off}} + k_p)C_0, \quad (\text{SI-3.1c})$$

$$\frac{dC_1}{dt} = k_pC_0 - (k_{\text{off}} + k_i)C_1, \quad (\text{SI-3.1d})$$

$$\frac{dC_2}{dt} = k_iC_1 - k_{\text{off}}C_2, \quad (\text{SI-3.1e})$$

$$\frac{dY}{dt} = \gamma_+^y(Y_T - Y) - \gamma_-^yY + \lambda C_1(Y_T - Y), \quad (\text{SI-3.1f})$$

$$\frac{dP}{dt} = \gamma_+^p(P_T - P) - \gamma_-^pP + \delta Y(P_T - P) - \mu C_1P. \quad (\text{SI-3.1g})$$

The state variables and parameters of the model (SI-3.1) are defined in (Lever et al., 2016). Parameters important for our derivation are: k_{on} and k_{off} are on- and off-rate constants, k_p is the kinetic proofreading rate constant, k_i is the kinetic rate constant for transforming of the active complex C_1 into the inactive complex C_2 . We will also need C_T , the total number of all ligand-receptor complexes,

$$C_T = C_0 + C_1 + C_2. \quad (\text{SI-3.2})$$

Here, C_T does not correspond to any conserved moiety and, instead, changes in time.

The model (SI-3.1) has the following first integrals, also termed moiety conservation relationships,

$$L_T = L + C_T. \quad (\text{SI-3.3a})$$

$$R_T = R + C_T, \quad (\text{SI-3.3b})$$

Due to the relationships (SI-3.3b) and (SI-3.3a), the corresponding first two equations (SI-3.1a) and (SI-3.1b) in the model (SI-3.1) become redundant and are omitted from further analysis.

Setting the model linearly independent equations (SI-3.1c) - (SI-3.1g) at steady state, we can obtain the following algebraic relationships,

$$C_0 = \left(\frac{k_{\text{on}}}{k_{\text{off}} + k_p} \right) \times LR, \quad (\text{SI-3.4a})$$

$$C_1 = \left(\frac{k_p}{k_{\text{off}} + k_i} \right) \times C_0, \quad (\text{SI-3.4b})$$

$$C_2 = \left(\frac{k_i}{k_{\text{off}}} \right) \times C_1, \quad (\text{SI-3.4c})$$

$$Y = \left(\frac{1 + (\lambda/\gamma_+^y) C_1}{1 + (\gamma_-^y/\gamma_+^y) + (\lambda/\gamma_+^y) C_1} \right) \times Y_T, \quad (\text{SI-3.4d})$$

$$P = \left(\frac{1 + (\delta/\gamma_+^p) Y}{1 + (\gamma_-^p/\gamma_+^p) + (\mu/\gamma_+^p) C_1 + (\delta/\gamma_+^p) Y} \right) \times P_T. \quad (\text{SI-3.4e})$$

Next, we eliminate the product LR from (SI-3.4a) by using (SI-3.4a) - (SI-3.4c) in (SI-3.2),

$$C_T = \left(\left(\frac{k_{\text{on}}}{k_{\text{off}} + k_p} \right) + \left(\frac{k_p}{k_{\text{off}} + k_i} \right) \left(\frac{k_{\text{on}}}{k_{\text{off}} + k_p} \right) + \left(\frac{k_i}{k_{\text{off}}} \right) \left(\frac{k_p}{k_{\text{off}} + k_i} \right) \left(\frac{k_{\text{on}}}{k_{\text{off}} + k_p} \right) \right) LR. \quad (\text{SI-3.5})$$

After simple algebraic manipulations, we obtain from (SI-3.5) that

$$LR = K_d C_T, \quad K_d = \frac{k_{\text{off}}}{k_{\text{on}}}. \quad (\text{SI-3.6})$$

Using (SI-3.6) in (SI-3.4a), and then (SI-3.4a) in (SI-3.4b), followed by using (SI-3.4b) in (SI-3.4c), we obtain

$$C_0 = \left(\frac{k_{\text{off}}}{k_{\text{off}} + k_p} \right) C_T, \quad (\text{SI-3.7a})$$

$$C_1 = \left(\frac{k_p}{k_{\text{off}} + k_i} \right) \left(\frac{k_{\text{off}}}{k_{\text{off}} + k_p} \right) C_T, \quad (\text{SI-3.7b})$$

$$C_2 = \left(\frac{k_i}{k_{\text{off}}} \right) \left(\frac{k_p}{k_{\text{off}} + k_i} \right) \left(\frac{k_{\text{off}}}{k_{\text{off}} + k_p} \right) C_T. \quad (\text{SI-3.7c})$$

Note that C_T is still unknown in (SI-3.7). To compute C_T , we use an alternative expression for the product LR .

Indeed, we can obtain from (SI-3.3a) and (SI-3.3b) that $L = L_T - C_T$ and $R = R_T - C_T$, respectively. Now, using $LR = (L_T - C_T)(R_T - C_T)$ in (SI-3.6), we come to a closed quadratic equation with respect to C_T ,

$$(L_T - C_T)(R_T - C_T) = K_d C_T. \quad (\text{SI-3.8})$$

Solving the quadratic equation (SI-3.8) with respect to C_T , we obtain two solutions, only one of which corresponds to the biologically meaningful condition, $C_T = 0$ at $L_T = 0$,

$$C_T = \frac{1}{2} \left(R_T + L_T + K_d - \sqrt{(R_T + L_T + K_d)^2 - 4R_T L_T} \right). \quad (\text{SI-3.9})$$

The solution (SI-3.9) also corresponds to the stable equilibrium in the system of linearly independent equations (SI-3.1c) - (SI-3.1g).

It is convenient to nondimensionalize the equilibrium solution of (SI-3.1) given by the expressions (SI-3.4a) - (SI-3.4e), and (SI-3.9) by scaling all state variables and parameters as follows,

$$c_T = \frac{C_T}{R_T}, \quad c_k = \frac{C_k}{R_T}, \quad k = 0, 1, 2, \quad (\text{SI-3.10a})$$

$$y = \frac{Y}{Y_T}, \quad u = \frac{P}{P_T}, \quad (\text{SI-3.10b})$$

$$K_P = \frac{1}{R_T} \left(\frac{k_P}{k_{\text{on}}} \right), \quad K_i = \frac{1}{R_T} \left(\frac{k_i}{k_{\text{on}}} \right), \quad (\text{SI-3.10c})$$

$$\Gamma^y = \frac{\gamma_-^y}{\gamma_+^y}, \quad \Gamma^p = \frac{\gamma_-^p}{\gamma_+^p}, \quad (\text{SI-3.10d})$$

$$\Lambda = \lambda \frac{R_T}{\gamma_+^y}, \quad \Delta = \delta \frac{Y_T}{\gamma_+^p}, \quad \Theta = \mu \frac{R_T}{\gamma_+^p}, \quad (\text{SI-3.10e})$$

$$\alpha = \frac{L_T}{R_T}, \quad \kappa = \frac{K_d}{R_T}. \quad (\text{SI-3.10f})$$

We obtain from the rescaled (SI-3.4e) that

$$u(\alpha, \kappa) = \frac{1 + \Delta y(\alpha, \kappa)}{1 + \Gamma^p + \Theta v(\kappa) c_T(\alpha, \kappa) + \Delta y(\alpha, \kappa)}. \quad (\text{SI-3.11})$$

In (SI-3.11), the functions $c_1(\alpha)$ and $y(\alpha)$ are obtained from the corresponding expressions (SI-3.4b) and (SI-3.4d) rescaled as discussed earlier,

$$y(\alpha, \kappa) = \frac{1 + \Lambda v(\kappa) c_T(\alpha, \kappa)}{1 + \Gamma^y + \Lambda v(\kappa) c_T(\alpha, \kappa)}, \quad (\text{SI-3.12a})$$

$$c_T(\alpha, \kappa) = \frac{1}{2} \left(1 + \alpha + \kappa - \sqrt{(1 + \alpha + \kappa)^2 - 4\alpha} \right), \quad (\text{SI-3.12b})$$

$$v(\kappa) = \left(\frac{K_P}{\kappa + K_i} \right) \left(\frac{\kappa}{\kappa + K_P} \right). \quad (\text{SI-3.12c})$$

Reference values of parameters used in the expressions (SI-3.11) - (SI-3.12) are listed in Table SI-3.1. These values correspond to the values used to compute Fig. 3 in (Lever et al., 2016).

Table SI-3.1. KPL-IFFL model parameter values.

Nº	Parameter	Value
1.	K_i	10^{-3}
2.	K_p	10^{-2}
3.	Γ^y	5×10^2
4.	Γ^p	5×10^2
5.	Δ	5×10^3
6.	Θ	5×10^4
7.	Λ	10^4
8.	α	$10^{-4} - 10^4$
9.	κ	$10^{-4} - 10^2$

SI-4 ANALYSIS OF SPARSE VERSUS DENSE EXPERIMENTAL DATA

The main limitation of experimental data (Kohlhapp et al., 2016) is that the data is sparse. Yet, in spite of this limitation, by focusing on the phenotypes (A) and (B) schematically depicted in Fig. 8, our model semi-quantitatively fits a body of experimental data both discussed in the current literature and in (Kohlhapp et al., 2016) with very a small number of variables and parameters.

The topic of limitations imposed by the sparsity of experimental data has been widely discussed in the biological and especially immunological literature in the context of the applicability of such data in mathematical modeling (De Boer and Perelson, 2013; François et al., 2013; Gunawardena, 2014; Eftimie et al., 2016) to mention just a few references.

Small-scale models are highly interpretative (James et al., 2013; Ledzewicz and Schattler, 2017) and, here, we agree with the following citation: “Simplified models are sometimes more predictive than elaborate ones when data are sparse and have the added benefit of transparency” (François et al., 2013). An added benefit of smaller and more phenomenological models is that they have a small number of parameters, for which one may be able to find rough estimates from the literature. In contrast, large scale models have many parameters, most of which may not be available from the literature and, instead, should be fitted to data. Such models may be more powerful in accurate predictions at the (possible) expense of losing interpretability (James et al., 2013).

SI-5 MATHEMATICAL AND NUMERICAL METHODS

The steady-state solutions of the models developed in this work, the solution stability (Sontag, 2013), as well as the parameter continuation of the steady-state solutions (Kuznetsov, 2013) have been studied numerically (Khibnik et al., 1993), using the command-line functionality of `matcont6p10`, a Matlab®-based Continuation Toolbox (Dhooge et al., 2008). MATLAB® Parallel Computing Toolbox was employed whenever possible. Finally, the color maps were generated using `varycolor.m`, a Matlab®-based function.

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